MDLM: Mining Deep Subspace Prior With Low-Rank Difference Minimization for Multispectral Pansharpening

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***Abstract*—Panchromatic sharpening (pansharpening) is a promising technology designed to fuse multispectral (MS) images with panchromatic (PAN) images. However, the low correlation between PAN and MS data often results in spectral and spatial inconsistencies, which limit the scalability of optimization-based models over multiple iterations. In this article, we propose a deep subspace (DS) prior-driven framework for MS pansharpening to fully leverage spatial–spectral priors. Unlike existing methods that rely solely on either data-driven or model-driven approaches, our work integrates the deep neural network's prior knowledge to guide the optimization process of an explicit pansharpening analytical model. Specifically, we propose a novel DS prior to address the problem of overcomplete representation and to maintain spatial–spectral consistency in a lightweight manner. In addition, we introduce a rank difference minimization (RDM) model that incorporates a nonlocal self-similar prior into the spatial fidelity term, effectively reducing interference and improving overall performance. To evaluate the proposed method, we conduct experiments on four mainstream satellite datasets, namely, WorldView-3 (WV3), WorldView-2 (WV2), QuickBird (QB), and IKONOS (IK), and further validate its generalization capability on GaoFen-2 (GF-2) scenes. The results demonstrate that our approach significantly enhances the spatial detail quality of fused images while reducing spectral distortion, thereby highlighting its effectiveness and potential for practical applications.**

***Index Terms*—Deep learning (DL), deep subspace (DS) prior, low-rank difference minimization (RDM), panchromatic sharpening (pansharpening).**

I. Introduction

With the advancement of satellite remote sensing technology, the availability of remote sensing images has significantly increased. Owing to the diverse requirements across various fields, remote sensing image processing has become an essential area of research, with image fusion emerging as a critical direction. Panchromatic sharpening (pansharpening) has become a pivotal image fusion technique that combines the rich spectral information of multispectral (MS) images with the detailed spatial information of panchromatic (PAN) images. This process generates high-resolution MS (HRMS) images, which are crucial for downstream tasks such as feature extraction [1], automatic identification [2], and target tracking [3]. However, fused images often suffer from spatial artifacts [4], chromatic aberrations [5], and image distortions [6], which pose significant challenges for subsequent applications.

Early pansharpening techniques relied on various well-established models, which can be broadly categorized into component substitution (CS) and multiresolution analysis (MRA) methods. CS-based approaches, such as principal component analysis (PCA) [7], Gram–Schmidt (GS) [8], and Brovey transform (BT) [9], transform MS images into a different spectral space, replace one of the components with the PAN image, and subsequently transform the data back to the original spectral space. While CS methods are computationally efficient and effectively preserve spatial details, they often introduce significant spectral distortions because of the inherently low correlation between PAN and MS images. To mitigate spectral distortions, MRA-based methods inject high-frequency information into downsampled MS images. Common MRA techniques include the á trous wavelet transform (ATWT) [10], [11], curvelet transform [12], Laplacian pyramid [13], and high-pass filtering (HPF) [14]. These approaches generally produce lower spectral distortions than CS-based methods do and provide greater flexibility in retaining spatial details. However, MRA methods are computationally intensive and may still face challenges in preserving

Received 12 November 2025; revised 5 March 2026 and 31 May 2026; accepted 22 June 2026. Date of publication 30 June 2026; date of current version 9 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62401292 and Grant 62476140, in part by the Postdoctoral Fellowship Program of China Postdoctoral Science Foundation under Grant GZC20240745, in part by Jiangsu Funding Program for Excellent Postdoctoral Talent under Grant 2024ZB682, and in part by the Natural Science Research Start-Up Foundation of Recruiting Talents of Nanjing University of Posts and Telecommunications under Grant NY225048. *(Corresponding authors: Hu Zhu; Lizhen Deng.)*

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Digital Object Identifier 10.1109/TGRS.2026.3708597

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fine spectral and spatial information because of the limitations of traditional multiscale decomposition techniques.

In recent years, deep learning (DL) has revolutionized numerous fields, including remote sensing image processing. Convolutional neural networks (CNNs) and other deep architectures have been extensively investigated for pansharpening because of their superior feature extraction capabilities. For example, the CNN architecture originally designed for super-resolution (SR) has been adapted for pansharpening. Feng et al. [15] introduced a computational imaging-based approach to achieve SR colouration, utilizing a model trained for SR colouration to perform pansharpening without the need for MS images. Similarly, Scarpa et al. [16] proposed a target-adaptive model that demonstrated robust performance even in cases of mismatch with the training set. However, most CNN-based methods directly input source images into the training network, which may not fully exploit fine-grained details, leading to a loss of high-frequency information after pansharpening.

Recent advancements have sought to address these shortcomings by developing deeper and more sophisticated network structures that incorporate mechanisms such as residual learning, attention modules, and generative adversarial networks (GANs) to enhance both spatial and spectral fidelity. For example, Zhou et al. [17] proposed a novel unsupervised GAN model for pansharpening that can be trained on full-size images without requiring ground truth (GT), thus constructing an entirely unsupervised framework. Despite these advancements, this approach does not fully leverage the information embedded in the original image. Other methods, such as Pan-Net [18] and Res-Net [19], [20], were designed to predict image details more effectively and preserve both spatial and spectral information. To enhance feature extraction, a crucial aspect of image fusion, Liang et al. [21] and Song et al. [22] extended the transformer mechanism to pixel-level fusion and learned regions of interest in low-resolution MS (LRMS) images by integrating CNNs within the framework. Tan et al. [23] approached the integration of spatial–frequency information from a hierarchical perspective, combining global and local frequency data while emphasizing the importance of global Fourier information and interbranch interactions. Wang et al. [24] proposed a two-stage network based on variational optimization (VO), leveraging prior information to estimate spatial–spectral degradation from the target image to the original images. Wang et al. [25] further enhanced the interpretability of image fusion through an intrinsically supervised deep unfolding network. He et al. [26] introduced Pan-Mamba, a state-space model that achieves efficient cross-modal information exchange and fusion through customized channel exchange and cross-modal Mamba modules. Despite these advancements, DL-based pansharpening methods continue to face several challenges, including significant spectral and spatial distortions resulting from the low correlation between PAN and MS images, as well as increasing computational complexity associated with deeper network architectures. Furthermore, many existing methods fail to effectively leverage the interactive information between PAN and MS images, which often leads to suboptimal fusion results.

This study is motivated by the subspace method, which is capable of removing significant Gaussian noise from images by transforming the problem into the estimation of subspace coefficients, thereby simplifying the estimation process. A simple and robust subspace learning method was introduced in [27], which demonstrated the potential of subspace representations to address noise-related challenges. Zhuang and Ng [28] extended the utility of subspace representations to address the problem of mixed noise. However, directly embedding a subspace from noisy data, as outlined in [27], results in the retention of fewer image components and leaves more spatial information within the learned subspace. Xu et al. [29] formulated two optimization problems regularized by a deep prior and generated corresponding models for PAN and LRMS images. These problems were subsequently solved using the gradient projection algorithm, with the iterative process extended across two network blocks. Low-rank optimization has also been applied to pansharpening [30], [31]; however, this approach struggles to effectively combine deep priors with simple linear estimations because of the two unmeasurable spaces between the optimization space and the deep knowledge space. Although high spatial resolution remote sensing data have become increasingly available in recent years, HRMS imagery is still expensive to acquire and often limited by sensor design, imaging conditions, and coverage constraints. In many practical remote sensing systems, PAN and MS observations remain a standard acquisition mode. Therefore, pansharpening is still of considerable practical significance. Moreover, the value of this work is not limited to spatial resolution enhancement alone, but also lies in improving spatial–spectral consistency and exploring a more effective integration of deep neural priors and model-based optimization. Based on the above discussion, the following questions are posed.

1) What prior knowledge can be utilized in the pansharpening task?
2) Where can this prior knowledge be applied within the pansharpening model?

To fully complement the spatial–spectral prior, we design a deep subspace (DS) prior-driven framework for pansharpening. The proposed method aims to leverage the prior knowledge embedded in neural networks through a subspace-based approach. Unlike existing methods that rely predominantly on pixel-based image priors [32], [33], our method uses DS representations to effectively transfer and seamlessly integrate deep knowledge into the pansharpening optimization model. This innovative approach addresses limitations in prior methods by enhancing the ability to embed richer spectral and spatial information. The differences between our proposed method and existing methods are illustrated in Fig. 1. The key contributions of our study are as follows.

1) To enable adaptive knowledge transfer through deep neural network embedding, we construct a subspace representation to guide the optimization of the deep prior constraint for the MS pansharpening problem, which can effectively mitigate the errors associated with direct pixel-level estimation.

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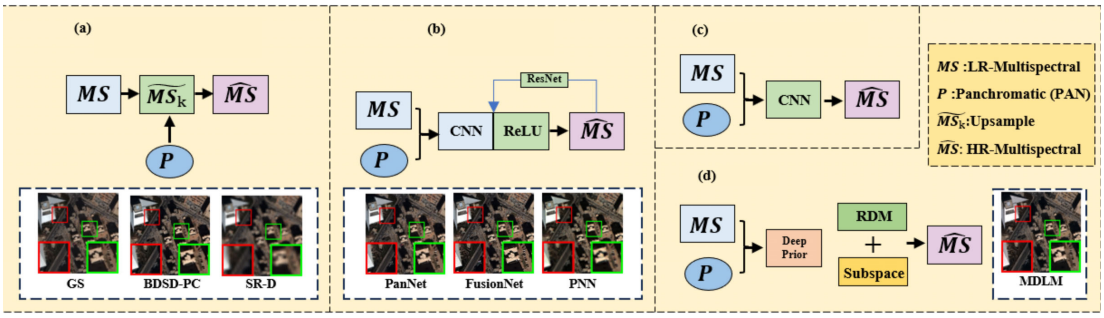


Fig. 1. Comparison of various pansharpening frameworks. (a) Classical methods. (b) DL methods. (c) Direct pixel estimation. (d) Our proposed MDLM with deep prior, RDM, and subspace representation.

2) The low-rank difference minimization (RDM) model is seamlessly integrated into our MDLM framework to reduce the interference caused by the low correlation between MS and PAN maps, thereby enhancing spatial fidelity.
3) Comprehensive experimental evaluations are conducted using four satellite datasets to assess the performance of the proposed framework. These evaluations compare our approach against a benchmark of state-of-the-art pansharpening methods.

The remainder of this article is organized as follows. Section II introduces the proposed MDLM method in detail. Section III discusses the optimization and update processes of the entire model, which are implemented using the alternating direction method of multipliers (ADMM). Section IV presents extensive experimental results, comparing the proposed approach with existing state-of-the-art methods. Finally, Section V concludes this article by summarizing the proposed MDLM framework and offering insights into potential future research directions.

II. Method

In this section, we first introduce the formulation of our proposed optimization model and then present the details of the framework.

A. Model Formulation

Definition 1 (Spectral Fidelity Term): Many existing spatial and spectral fusion methods upsample LRMS images to the PAN scale. However, the performance of such interpolation processes is often compromised by the introduction of inaccurate image information. To address this issue, our algorithm uses a downsampled version of the ideal HRMS image (HR-MSI), referencing the point spread function (PSF) of the MS sensor to design the blur kernel. Consequently, we model the observed imaging process as follows:

$$\mathbf{Y} = \mathbf{XBS} + \xi_1 \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{HW \times HW}$ represents the matrix form of the blurring operator, $\mathbf{S} \in \mathbb{R}^{HW \times hw}$ denotes the sampling matrix containing sparse components, and ξ_1 corresponds to Gaussian noise. The spectral fidelity term is then expressed as

$$f_{\text{spec}}\ (\mathbf{X}, \mathbf{Y}) = \|\mathbf{XBS} - \mathbf{Y}\|_F^2 \tag{2}$$

where $\|\cdot\|_{\text{F}}^2$ denotes the Frobenius norm.

Definition 2 (Spatial Fidelity Term): To increase the fidelity of spatial details during pansharpening, this article defines the following fidelity term formula based on the MRA method proposed by Vivone et al. [34]. The spatial fidelity term aims to improve spatial details while preserving spectral information. The general MRA framework is expressed as follows:

$$\mathbf{X}^{(\text{i})} = \mathbf{Y}_{\text{H}}^{(\text{i})} + \mathbf{G}^{(\text{i})} \circ \left(\widehat{\mathbf{P}}^{(\text{i})} - \widehat{\mathbf{P}}_{\text{L}}^{(\text{i})}\right). \tag{3}$$

For $i = 1, 2, \ldots, S$, where $\mathbf{Y}_{\text{H}}^{(\text{i})}, \mathbf{G}^{(\text{i})} \in \mathbb{R}^{H \times W}$ denotes the upsampled version of $\mathbf{Y}(i)$ on the scale of $\widehat{\mathbf{P}}^{(\text{i})}$ and the injection coefficient, respectively; $\circ$ denotes elementwise multiplication; and $\widehat{\mathbf{P}}_{\text{L}}^{(\text{i})}$ is a low-pass version of $\widehat{\mathbf{P}}^{(\text{i})}$, filtered by the impulse response h_i, i.e., $\widehat{\mathbf{P}}_{\text{L}}^{(\text{i})} = \widehat{\mathbf{P}}^{(\text{i})} * h_i$, where $*$ denotes the convolution operation. In addition, $\widehat{\mathbf{P}}^{(\text{i})} - \widehat{\mathbf{P}}_{\text{L}}^{(\text{i})}$ are treated as image details of the high-pass version of $\widehat{\mathbf{P}}^{(\text{i})}$. Two common options for defining the injection gain are $\mathbf{G}^{(\text{i})} = 1$, the so-called additive injection scheme, and $\mathbf{G}^{(\text{i})} = \mathbf{Y}_{\text{H}}^{(\text{i})} \oslash \widehat{\mathbf{P}}_{\text{L}}^{(\text{i})}$, referred to as the multiplying injection scheme or high-pass modulation (HPM) scheme, where $\oslash$ denotes elementwise division. The latter scheme, which is adopted in this section, effectively reproduces the local intensity contrast of PAN images within HR-MSIs and is preferred over the former because of its superior flexibility in configuring local weights.

To avoid introducing the LRMS image $\mathbf{Y}$, this scheme replaces $\mathbf{Y}_{\text{H}}$ with $\mathbf{XB}$, thereby enabling independent modeling of the relationship between the unknown HR-MSI and the PAN image. In matrix form, this relationship can be expressed as

$$\mathbf{G} \circ \left(\widehat{\mathbf{P}} - \widehat{\mathbf{P}}_{\text{L}}\right) = \mathbf{X} - \mathbf{XB} + \xi_2 \tag{4}$$

where ξ_2 represents the error term, which is assumed to follow a zero-mean normal distribution $N(0, \sigma^2)$ with a standard deviation of σ^2. When applied to histogram-matched PAN images, this implicit formulation facilitates mapping from an unknown HR-MSI to detailed images that are closely aligned with the PAN images. Notably, compared with the structural similarity model based on a linear combination assumption, the spatial fidelity model expressed in the above equation is capable of retaining spectral content to the maximum extent while simultaneously enhancing spatial details. Furthermore, this approach effectively leverages the dual properties of elementwise mapping derived from MRA and the high-frequency domain, as implemented through the HPM injection model.

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Therefore, the MRA-based detail injection spatial fidelity model can be expressed as follows:

$$f_{\text{spat}}\ (\mathbf{X},\mathbf{P}) = \left\| \mathbf{X} - \left(\mathbf{XB} + \mathbf{G} \circ \left(\widehat{\mathbf{P}} - \widehat{\mathbf{P}}_{\text{L}} \right) \right) \right\|_{\text{F}}^{2}. \tag{5}$$

In this article, we adopt the HPM scheme to perform detail extraction. Specifically, we define $\mathbf{G} = \mathbf{XB} \oslash \widehat{\mathbf{P}}\text{L}$. By directly using $\mathbf{XB}$ instead of $\mathbf{Y}\text{H}$, we derive our formulation as follows:

$$\mathbf{XB} + \mathbf{G} \circ \left(\widehat{\mathbf{P}} - \widehat{\mathbf{P}}_{\text{L}} \right) = \mathbf{XB} \circ \left(\widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_{\text{L}} \right) \tag{6}$$

where $\widehat{\mathbf{P}}_{\text{L}}$ is defined using an MTF-based filter. The final spatial fidelity term, which is dependent on the HPM scheme, can be formalized as follows:

$$f_{spat}\left(\mathbf{X},\mathbf{P}\right) = \left\| \mathbf{X} - \mathbf{XB} \circ \left(\widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_{\text{L}} \right) \right\|_{\text{F}}^{2}. \tag{7}$$

Definition 3 (Deep Prior): We introduce a deep prior that aims to integrate the output of CNN-based methods into the VO framework. The deep prior establishes a relationship between the output and the target HR-MSI, thereby transforming the prior information derived from CNNs into a variational pansharpening framework for further optimization. In this model, deep priors serve as general regularizers, eliminating the need for other prior information (e.g., low-rank and sparse priors) commonly employed in the literature. In this section, we directly input the observed images into the CNN-based deep network and use the output of the deep network to constrain the prior of our pansharpening model, as follows:

$$DP = f\left(\mathbf{X}, \mathbf{X}_{net}\right) \tag{8}$$

where DP denotes the deep prior, $f(\cdot)$ represents the mapping relationship established by the network, and $\mathbf{X}_{net}$ represents the output of the CNN.

Remark 1: The significant difference between the RDM model and existing low-rank methods (such as nuclear norm minimization and weighted nuclear norm minimization) lies in the fact that we analyze the rank minimization problem from the perspective of mathematical approximation theory. Specifically, the RDM model addresses the rank minimization problem by minimizing the rank difference, enabling the singular values of the recovered matrix to progressively align with those of the reference matrix. In the low-rank matrix recovery process, we aim for the singular values of the recovered matrix $\mathbf{X}^*$ to approximate those of the original matrix as closely as possible. To formalize this aim, we define the rank difference as follows:

$$\gamma^* = \sigma - \psi^* \tag{9}$$

where σ and ψ^* represent the singular values of $\mathbf{Y}$ and $\mathbf{X}^*$, respectively. The estimation accuracy of $\mathbf{X}^*$ is largely determined by this rank residual. However, in practical applications, the original matrix $\mathbf{X}^*$ is often unattainable, so we aim to estimate $\mathbf{X}^*$ using a good approximation $\mathbf{X}'$. By introducing $\mathbf{X}'$, we define the rank difference as

$$\gamma = \sigma - \psi \tag{10}$$

where ψ represents the singular values of $\mathbf{X}'$.

To estimate $\mathbf{X}$, this article draws inspiration from the self-similarity structures commonly found in natural images. Specifically, we directly select similar patches from the observed image to form a local neighborhood for each given patch (each patch represents a block of the image). By using the self-similarity method, we can then estimate the matrix group $\mathbf{X}_i$. The well-known nonlocal self-similarity (NSS) learning first considered the redundancy of natural image textures and structures in nonlocal regions. In other words, given a reference patch, one can search for multiple similar patches in the image. Formally, an observed image $y \in \mathbb{R}^N$ is divided into overlapping patches of size $\sqrt{d} \times \sqrt{d}$. Each patch is represented as $y_i \in \mathbb{R}^d$, where $i = 1, 2, \ldots, n$. For the ith patch $\mathbf{y}_i$, we select m similar patches from a surrounding window of $L \times L$ pixels (used for searching), forming the set $\mathbf{S}_i$. These selected patches are then stacked into a matrix $\mathbf{Y}_i \in \mathbb{R}^{d \times m}$, i.e., $\mathbf{Y}_i = \{y_{i1}, y_{i2}, \ldots, y_{im}\}$. Thus, the matrix $\mathbf{Y}_i$, composed of patches with similar structures, is considered a group, where $\{\mathbf{y}_{i,j}\}$ represents the jth patch in group i. We then define $\mathbf{Y}_i = \mathbf{X}_i + \mathbf{N}_i$, where $\mathbf{X}_i$ and $\mathbf{N}_i$ represent the original image and the noise counterpart matrices, respectively. Since all patches in each matrix group share similar structures, the matrix formed by the group exhibits low-rank properties. By solving this optimization problem, we can estimate the low-rank matrix $\mathbf{X}_i$ as follows:

$$\widehat{\mathbf{X}}_i = \arg\min_{\mathbf{X}_i} \frac{1}{2} \left\| \mathbf{Y}_i - \mathbf{X}_i \right\|_F^2 + \lambda \left\| \gamma_i \right\|_1 \tag{11}$$

where $\gamma_i = \sigma_i - \psi_i$, with σ_i and ψ_i representing the singular values of $\mathbf{Y}_i$ and $\mathbf{X}_i$, respectively. $\mathbf{X}_i \in \mathbb{R}^{d \times m}$ is a good estimate of the patch group $\mathbf{X}_i^*$ from the original image.

Here, $\|\gamma\|_1$ denotes the aggregation of rank-difference residues over all nonlocal patch groups, i.e., $\|\gamma\|_1 = \sum_i \|\gamma_i\|_1$.

The objective function of the RDM algorithm is formulated based on the observation model and the rank difference prior. This formulation can be expressed as the following convex optimization problem:

$$\min_{\mathbf{X}} \frac{1}{2} f_{spec}\left(\mathbf{X},\mathbf{Y}\right) + \lambda f_{spat}\left(\mathbf{X},\mathbf{P}\right) + \tau \|\gamma\|_1. \tag{12}$$

Remark 2: To enhance the accuracy, generalization, and efficiency of the pansharpening algorithm, this method introduces a subspace representation based on the RDM model to guide the constrained optimization of the deep prior. The shared subspace facilitates knowledge transfer, enabling the formation of DS embedding. This approach effectively mitigates the "error of direct pixel-level estimation", which arises when standard CNNs neglect intrinsic spectral correlations during pixel-wise regression. Consequently, the proposed method allows for robust processing of large-scale and high-dimensional MS data. The optimization model is expressed as follows:

$$\min_{\mathbf{X}} f_{VO}\left(\mathbf{X},\mathbf{Y},\mathbf{P}\right) + \alpha f_{DS}\left(\mathbf{X},\mathbf{X}_{net},\mathbf{EZ}\right) \tag{13}$$

where $f_{VO}(\mathbf{X},\mathbf{Y},\mathbf{P})$ includes the recovered image $\mathbf{X}$ (HRMS), the observed image $\mathbf{Y}$ (LRMS image) and $\mathbf{P}$ (PAN image) and where $\mathbf{Y}$ is the MS image, and $\mathbf{P}$ is the PAN image. $f_{DS}(\mathbf{X},\mathbf{X}_{net})$ represents the DS embedding term, which plays a guiding role, where α is the balancing parameter between the VO and DS terms. $\mathbf{X}_{net} \in \mathbb{R}^{H \times W \times S}$ represents the output of the general CNN method; $\mathbf{EZ} = \mathbf{X}$ is subspace notation;

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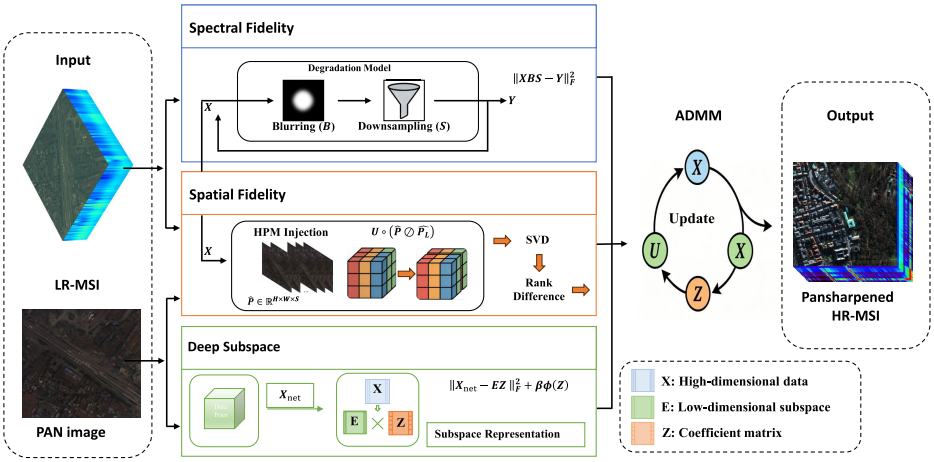


Fig. 2. Overview of the proposed MDLM framework. The method integrates three key components: 1) spectral fidelity for degradation modeling, 2) spatial fidelity utilizing RDM and HPM injection, and 3) DS for embedding CNN priors into a low-dimensional subspace. The entire model is solved via an ADMM algorithm to iteratively reconstruct the HR-MSI.

$\mathbf{E} \in \mathbb{R}^{H\times W\times S}$ represents the subspace basis for the spatial representation of $\mathbf{X}$; $\mathbf{Z} \in \mathbb{R}^{K\times W\times S}$ represents the coefficient of $\mathbf{X}$ with respect to the basis $\mathbf{E}$; and $\mathbf{EZ}$ refers to the operation, where $\mathbf{Z}$ is multiplied with $\mathbf{E}$ along the third dimension (i.e., $\mathbf{E}^{(i)}\mathbf{Z}^{(i)} \in \mathbb{R}^{H\times W}$, $\mathbf{EZ} \in \mathbb{R}^{H\times W\times S}$).

In the context of pansharpening, directly applying linear constraints to incorporate prior information from DL may not effectively enforce the desired prior constraints. This is due to the differences in distributions between the two immeasurable spaces, which limit the effectiveness of the approach. As a result, the DL prior information cannot be effectively integrated with the deep prior through simple linear estimation. To address this issue, this article introduces a subspace representation to guide the constrained optimization of the deep prior. The shared subspace between the two distributions facilitates knowledge transfer, thereby achieving DS embedding.

In Wu's study [35], the following proximal deep injection terms were introduced:

$$f_{PDI}\left(\mathbf{X}, \mathbf{X}_{net}\right) = \left\|\mathbf{X} - \mathbf{X}_{net}\right\|_{\mathrm{F}}^{2}. \tag{14}$$

By using this proximal term, we aim to integrate the output of CNN-based methods into the VO framework. We introduce a DS embedding term that establishes a relationship between the output and the target HR-MSI, thereby transforming the prior information required by the CNN into a variational pansharpening framework for further optimization. In this model, the DS embedding term functions as a regularizer. Given the large size of HRMS data and the high correlation between channels, we integrate the DS embedding term into a subspace representation approach. This method converts the large amount of HRMS data into the product of the subspace basis $\mathbf{E}$ and its representation coefficient $\mathbf{Z}$; at the same time, the subspace basis $\mathbf{E}$ can be learned from observations. The DS embedding term is defined as follows:

$$f_{DS}\left(\mathbf{X}, \mathbf{X}_{net}, \mathbf{EZ}\right) = \left\|\mathbf{X}_{net} - \mathbf{EZ}\right\|_{\mathrm{F}}^{2} + \beta\phi\left(\mathbf{Z}\right) \tag{15}$$

where β is a positive parameter that balances the VO and DS terms, and $\phi(\cdot)$ is a regularization term that encodes prior information specific to the image.

Based on the observation model and neural network prior, the objective function of the proposed algorithm is designed to formulate the pansharpening problem as a convex optimization task. By incorporating the low-RDM and the DS embedding term introduced in this work, shown in Fig. 2 the final form of the proposed model can be expressed as follows:

$$\begin{aligned}\min_{\mathbf{X},\mathbf{Z}} \frac{1}{2}\left\|\mathbf{XBS} - \mathbf{Y}\right\|_{\mathrm{F}}^{2} + \lambda\left\|\mathbf{X} - \mathbf{XB}\circ\left(\widehat{\mathbf{P}}\oslash\widehat{\mathbf{P}}_{\mathrm{L}}\right)\right\|_{\mathrm{F}}^{2} + \tau\left\|\gamma\right\|_{1} \\ + \alpha\left(\left\|\mathbf{X}_{net} - \mathbf{EZ}\right\|_{\mathrm{F}}^{2} + \beta\phi\left(\mathbf{Z}\right)\right).\end{aligned} \tag{16}$$

B. Algorithm Optimization

The abovementioned optimization problem of the DS embedding pansharpening algorithm under low-rank difference is challenging to solve directly using gradient descent because of the multiple variables and constraints. In this article, an algorithm based on the ADMM is proposed. To solve the formulated mixed problem, several auxiliary variables are introduced to represent the constraints of the pansharpening model. Finally, it is divided into multiple subproblems, which are then solved iteratively. Specifically, the auxiliary variables $\mathbf{U} = \mathbf{XB}$ and $\mathbf{EZ} = \mathbf{X}$ are introduced, transforming the original model into the following form:

$$\begin{aligned}\min_{\mathbf{X},\mathbf{U},\mathbf{Z}} \frac{1}{2}\left\|\mathbf{US} - \mathbf{Y}\right\|_{F}^{2} + \lambda\left(\left\|\mathbf{X} - \mathbf{U}\circ\left(\widehat{\mathbf{P}}\oslash\widehat{\mathbf{P}}_{\mathbf{L}}\right)\right\|_{F}^{2}\right. \\ \left. + \tau\left\|\gamma\right\|_{1}\right) + \alpha\left(\left\|\mathbf{X}_{net} - \mathbf{EZ}\right\|_{F}^{2} + \beta\phi\left(\mathbf{Z}\right)\right) \\ \text{s.t.} \quad \mathbf{U} = \mathbf{XB}, \mathbf{EZ} = \mathbf{X}.\end{aligned} \tag{17}$$

Considering that the above optimization problem is a constrained optimization problem, we utilize the augmented Lagrangian function to transform this constrained optimization problem into an unconstrained optimization problem

$$L_{\eta_1,\eta_2}\left(\mathbf{X}, \mathbf{U}, \mathbf{Z}, \Lambda_1, \Lambda_2\right) = \frac{1}{2}\left\|\mathbf{US} - \mathbf{Y}\right\|_{F}^{2} + \alpha\left(\left\|\mathbf{X}_{net} - \mathbf{EZ}\right\|_{F}^{2}\right.$$

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$$+\beta\phi(\mathbf{Z})\big)+\lambda\left(\left\|\mathbf{X}-\mathbf{U}\circ\left(\widehat{\mathbf{P}}\oslash\widehat{\mathbf{P}}_{\mathbf{L}}\right)\right\|_F^2+\tau\|\gamma\|_1\right)+\frac{\eta_1}{2}\|\mathbf{XB}-\mathbf{U}+\Lambda_1/\eta_1\|_F^2+\frac{\eta_2}{2}\|\mathbf{X}-\mathbf{EZ}+\Lambda_2/\eta_2\|_F^2 \quad (18)$$

where $\gamma=\sigma-\psi$, σ, ψ represent the singular values of $\mathbf{X}$ and $\mathbf{X}'$ (an estimate of $\mathbf{X}$), respectively. λ is a positive regularization parameter. $\mathbf{EZ}=\mathbf{X}$ represents the subspace representation of $\mathbf{X}$; and $\phi(\cdot)$ is a regularization term that can represent prior information based on the image structure. Λ_1 and Λ_2 are Lagrange multipliers, and η_1 and η_2 are positive multiplication factors.

By fixing other variables while updating the current variable, we solve the above formula through alternating iterative methods. This process can be understood in terms of solving the following subproblems iteratively:

$$\begin{aligned}\mathbf{X}^{i+1}&\in\arg\min_{\mathbf{X}}L_{\eta_1,\eta_2}\left(\mathbf{X},\mathbf{U}^i,\mathbf{Z}^i,\Lambda_1^i,\Lambda_2^i\right)\\ \mathbf{U}^{i+1}&\in\arg\min_{\mathbf{U}}L_{\eta_1,\eta_2}\left(\mathbf{X}^{i+1},\mathbf{U},\mathbf{Z}^i,\Lambda_1^i,\Lambda_2^i\right)\\ \mathbf{Z}^{i+1}&\in\arg\min_{\mathbf{Z}}L_{\eta_1,\eta_2}\left(\mathbf{X}^{i+1},\mathbf{U}^{i+1},\mathbf{Z},\Lambda_1^i,\Lambda_2^i\right).\end{aligned} \quad (19)$$

III. ADMM Algorithm

A. Update X

Update the variable $\mathbf{X}$. The subproblem involving $\mathbf{X}$ in the formulation is as follows:

$$\arg\min_{\mathbf{X}}\lambda\left\|\mathbf{X}-\mathbf{U}\circ\left(\widehat{\mathbf{P}}\oslash\widehat{\mathbf{P}}_{\mathrm{L}}\right)\right\|_F^2+\frac{\eta_1}{2}\left\|\mathbf{XB}-\mathbf{U}+\frac{\Lambda_1}{\eta_1}\right\|_F^2+\frac{\eta_2}{2}\left\|\mathbf{X}-\mathbf{EZ}+\frac{\Lambda_2}{\eta_2}\right\|_F^2+\lambda\tau\|\gamma\|_1. \quad (20)$$

The first three terms of the above equation are differentiable convex functions, and the gradient satisfies the L-Lipschitz condition, while the term $\tau\|\gamma\|_1$ is clearly a nonsmooth function. Thus, we introduce the proximal gradient descent (PGD) method [36] to solve for $\mathbf{X}$. Compared with the classical and stochastic gradient descent methods, the proximal gradient descent method has a narrower application scope. Specifically, for convex optimization problems, PGD is applicable when the objective function consists of both a differentiable convex part and a nondifferentiable (or locally nondifferentiable) convex part. The objective function can be written as the sum of two functions, $f(x)$ and $g(x)$, where $f(x)$ is a differentiable convex function, and $g(x)$ is a nondifferentiable (or locally nondifferentiable) convex function. The above formula can be rewritten as

$$\arg\min_{\mathbf{X}}f(\mathbf{X})+\lambda\tau g(\mathbf{X}) \quad (21)$$

$$f(\mathbf{X})=\lambda\left\|\mathbf{X}-\mathbf{U}\circ\left(\widehat{\mathbf{P}}\oslash\widehat{\mathbf{P}}_{\mathrm{L}}\right)\right\|_F^2+\frac{\eta_1}{2}\left\|\mathbf{XB}-\mathbf{U}+\frac{\Lambda_1}{\eta_1}\right\|_F^2+\frac{\eta_2}{2}\left\|\mathbf{X}-\mathbf{EZ}+\frac{\Lambda_2}{\eta_2}\right\|_F^2 \quad (22)$$

$$g(\mathbf{X})=\|\gamma\|_1. \quad (23)$$

There is also

$$\nabla f(\mathbf{X})=2\lambda\left(\mathbf{X}-\mathbf{U}\circ\left(\widehat{\mathbf{P}}\oslash\widehat{\mathbf{P}}_{\mathrm{L}}\right)\right)+\eta_1\left(\mathbf{XBB}^T-\mathbf{UB}^T\right)+\Lambda_1\mathbf{B}^T+\eta_2\left(\mathbf{X}-\mathbf{EZ}+\frac{\Lambda_2}{\eta_2}\right). \quad (24)$$

The PGD algorithm [36] is as follows. Let $\mathbf{X}^K$ be the result after the update of $\mathbf{X}$. For $\mathbf{X}\to\mathbf{X}^K$, the approximate value of $f(\mathbf{X})$ can be expressed as

$$\begin{aligned}\widehat{f}\left(\mathbf{X},\mathbf{X}^k\right)&=f\left(\mathbf{X}^k\right)+\left\langle\nabla f^T\left(\mathbf{X}^k\right),\left(\mathbf{X}-\mathbf{X}^k\right)\right\rangle+\frac{L}{2}\left\|\mathbf{X}-\mathbf{X}^k\right\|_F^2\\&=\frac{L}{2}\left\|\mathbf{X}-\left(\mathbf{X}^k-\frac{1}{L}\nabla f\left(\mathbf{X}^k\right)\right)\right\|_F^2\end{aligned} \quad (25)$$

where $L\geqslant 1$, and $1/L$ represents the step size; therefore, the result of the $\mathbf{X}^{k+1}$ iteration is as follows:

$$\begin{aligned}\mathbf{X}^{k+1}&=\arg\min_{\mathbf{X}}f(\mathbf{X})+\lambda\tau g(\mathbf{X})\\&=\arg\min_{\mathbf{X}}\widehat{f}\left(\mathbf{X},\mathbf{X}^k\right)+\lambda\tau g(\mathbf{X})\\&=\arg\min_{\mathbf{X}}\frac{L}{2}\left\|\mathbf{X}-\left(\mathbf{X}^k-\frac{1}{L}\nabla f\left(\mathbf{X}^k\right)\right)\right\|_F^2+\lambda\tau g(\mathbf{X})\\&=\arg\min_{\mathbf{X}}\frac{1}{2}\left\|\mathbf{X}-\mathbf{Q}\right\|_F^2+\frac{\lambda\tau}{L}\|\gamma\|_1.\end{aligned} \quad (26)$$

To solve the minimization problem in (26), we analyze the rank difference term in the singular value domain. The goal is to align the spectral components of the estimated image with those of the reference prior.

The minimization problem in (26) can be efficiently solved using a singular value decomposition (SVD)-based approach combined with the proposed RDM. Before applying SVD, the intermediate variable $\mathbf{Q}$ is unfolded into a matrix form along the spectral mode. First, we perform SVD on the intermediate variable $\mathbf{Q}$

$$\mathbf{Q}=\mathbf{L}\Sigma\mathbf{R}^T \quad (27)$$

where $\mathbf{L}$ and $\mathbf{R}$ denote the left and right singular vector matrices, respectively, and $\Sigma=\mathrm{diag}(\xi_1,\xi_2,\ldots,\xi_n)$ represents the singular value matrix of the current estimate. We specifically employ the notation $\mathbf{L}$ and $\mathbf{R}$ to distinguish the singular vectors from the auxiliary variable $\mathbf{U}$ used in the ADMM framework.

Simultaneously, we derive the reference singular values from the reference low-rank matrix $\mathbf{X}'$ [27], which is constructed by stacking nonlocal similar patches. Its SVD is expressed as $\mathbf{X}'=\mathbf{L}_{ref}\mathbf{\Phi}\mathbf{R}_{ref}^T$, where $\mathbf{\Phi}=\mathrm{diag}(\varphi_1,\varphi_2,\ldots,\varphi_n)$ contains the reference singular values capturing the structural correlation.

The core optimization is achieved via a *shifted soft-thresholding* operation. Unlike standard nuclear norm minimization that shrinks singular values toward zero, we shrink the deviation between the observed values ξ_i and the reference values φ_i

$$\sigma_i=\mathcal{S}_{\tau'}(\xi_i-\varphi_i)+\varphi_i \quad (28)$$

where $\mathcal{S}_{\tau'}(\cdot)$ denotes the soft-thresholding operator with threshold $\tau'=\lambda\tau/L$. This operator effectively minimizes the rank difference residue, ensuring that the reconstructed image structure progressively approximates the reference structural information while suppressing noise.

Finally, the optimized singular value matrix is formed as $\mathbf{\Omega}=\mathrm{diag}(\sigma_i)$, and the updated estimate is reconstructed using the computed bases $\mathbf{L}$ and $\mathbf{R}$

$$\mathbf{X}^{k+1}=\mathbf{L}\mathbf{\Omega}\mathbf{R}^T. \quad (29)$$

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Algorithm 1 Proposed MDLM Algorithm

Input: LR-MSI $\mathbf{Y}$, PAN $\mathbf{P}$, CNN output $\mathbf{X}_{net}$, Blur $\mathbf{B}$, Downsampler $\mathbf{S}$.

Output: High-resolution MS image $\mathbf{X}$.

Initialization:
  Initialize $\mathbf{X}$ by upsampling $\mathbf{Y}$.
  Compute basis $\mathbf{E}$ from $\mathbf{X}_{net}$ via SVD.
  Initialize auxiliaries $\mathbf{U}, \mathbf{Z}, \Lambda_1, \Lambda_2$ to $\mathbf{0}$.
  Pre-compute HPM term: $\mathbf{M}_{HPM} \leftarrow \widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_L$.
while not converged **do**
Step 1: Update X
  $\mathbf{Q} \leftarrow \mathbf{X}^k - \frac{1}{L}\nabla f(\mathbf{X}^k)$
  $[\mathbf{L}, \mathbf{\Sigma}, \mathbf{R}] \leftarrow \text{SVD}(\mathbf{Q})$
  $\mathbf{\Omega} \leftarrow \mathcal{S}_{\tau'}(\mathbf{\Sigma} - \mathbf{\Phi}) + \mathbf{\Phi}$
  $\mathbf{X} \leftarrow \mathbf{L}\mathbf{\Omega}\mathbf{R}^T$
Step 2: Update U
  $\mathbf{U} \leftarrow \dfrac{\mathbf{Y}\mathbf{S}^T + \eta_1\mathbf{X}\mathbf{B} + 2\lambda\mathbf{X} \circ \mathbf{M}_{HPM} + \Lambda_1}{\mathbf{D}_{SS^T} + \eta_1\mathbf{1} + 2\lambda\mathbf{M}_{HPM} \circ \mathbf{M}_{HPM}}$
Step 3: Update Z
  $\mathbf{Z} \leftarrow \mathcal{D}_\sigma\left(\mathbf{E}^T\left(\dfrac{2\alpha\mathbf{X}_{net} + \eta_2(\mathbf{X} + \Lambda_2/\eta_2)}{2\alpha + \eta_2}\right)\right)$
Step 4: Update Multipliers
  $\Lambda_1 \leftarrow \Lambda_1 + \eta_1(\mathbf{X}\mathbf{B} - \mathbf{U})$
  $\Lambda_2 \leftarrow \Lambda_2 + \eta_2(\mathbf{X} - \mathbf{E}\mathbf{Z})$
  Check Convergence: $\|\mathbf{X}_{new} - \mathbf{X}_{old}\|_F / \|\mathbf{X}_{old}\|_F \le \epsilon$
end while
return X

This procedure corresponds to the **Update X** step in Algorithm 1. Notably, in each iteration, the reference matrix $\mathbf{X}'$ is updated via nonlocal patch matching, aligning the singular values of $\mathbf{X}$ with the reference to preserve structural consistency.

B. Update U

Update the variable $\mathbf{U}$. The subproblem involving $\mathbf{U}$ in the formula is as follows:

$$\arg\min_{\mathbf{U}} \frac{1}{2}\|\mathbf{U}\mathbf{S} - \mathbf{Y}\|_F^2 + \lambda\left\|\mathbf{X} - \mathbf{U} \circ \left(\widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_{\mathbf{L}}\right)\right\|_F^2 + \frac{\eta_1}{2}\left\|\mathbf{X}\mathbf{B} - \mathbf{U} + \frac{\Lambda_1}{\eta_1}\right\|_F^2. \tag{30}$$

Given the scale factor r, the extraction mask is defined as a sparse matrix $\mathbf{K} \in \mathbb{R}^{r\times r}$, where the entries are set to 1 only at a position determined by the extraction scheme, e.g., when $r = 4$, $\mathbf{K}[(3,3)] = 1$. For all other cases, $\mathbf{K}[l]$ is 0. The following formula is thus obtained:

$$\mathbf{U}\mathbf{S}\mathbf{S}^T = \mathbf{U} \circ \mathbf{D}_{SS^T} \tag{31}$$

where $\mathbf{D}_{SS^T} = [\text{vec}(\mathbf{K} \otimes 1), \text{vec}(\mathbf{K} \otimes 1), \ldots, \text{vec}(\mathbf{K} \otimes 1)]^T$, $\bigotimes$ represents the Kronecker product, and $\text{vec}(\cdot)$ represents the vectorization operator, $ł \in \mathbb{R}^{h\times w}$. According to the above formula, we can deduce the following:

$$\mathbf{U} = \frac{\mathbf{Y}\mathbf{S}^T + \eta_1\mathbf{X}\mathbf{B} + 2\lambda\mathbf{X} \circ \left(\widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_{\mathrm{L}}\right) + \Lambda_1}{\mathbf{D}_{SS^T} + \eta_1 1 + 2\lambda\left(\widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_{\mathrm{L}}\right) \circ \left(\widehat{\mathbf{P}} \oslash \widehat{\mathbf{P}}_{\mathrm{L}}\right)}. \tag{32}$$

Notably, this formula also needs to be divided elementwise when solving.

C. Update Z

Update the variable $\mathbf{Z}$. The subproblem involving $\mathbf{Z}$ in the formula is as follows:

$$\arg\min_{\mathbf{Z}} \alpha\left(\|\mathbf{X}_{net} - \mathbf{E}\mathbf{Z}\|_F^2 + \beta\phi(\mathbf{Z})\right) + \frac{\eta_2}{2}\left\|\mathbf{X} - \mathbf{E}\mathbf{Z} + \frac{\Lambda_2}{\eta_2}\right\|_F^2. \tag{33}$$

Expanding the contents within the parentheses of the above formula yields the following:

$$\arg\min_{\mathbf{Z}} \alpha\beta\phi(\mathbf{Z}) + \alpha\|\mathbf{X}_{net} - \mathbf{E}\mathbf{Z}\|_F^2 + \frac{\eta_2}{2}\left\|\mathbf{X} - \mathbf{E}\mathbf{Z} + \frac{\Lambda_2}{\eta_2}\right\|_F^2. \tag{34}$$

According to the definition and operation rules of the F norm, the above formula can be simplified as

$$\arg\min_{\mathbf{Z}} \alpha\beta\phi(\mathbf{Z}) + \left(\frac{2\alpha + \eta_2}{2}\right)\left\|\frac{1}{2\alpha + \eta_2}\left(2\alpha\mathbf{X}_{net} + \eta_2\left(\mathbf{X} + \frac{\Lambda_2}{\eta_2}\right)\right) - \mathbf{E}\mathbf{Z}\right\|_F^2. \tag{35}$$

Since $\mathbf{E}$ is a semiorthogonal matrix, for all $x \in \mathbb{R}^p$, there is an equidistant property. Thus, we can derive the following:

$$\arg\min_{\mathbf{Z}} \frac{\alpha\beta}{2\alpha + \eta_2}\phi(\mathbf{Z}) + \frac{1}{2}\left\|\mathbf{E}^T\left(\frac{2\alpha}{2\alpha + \eta_2}\mathbf{X}_{net} + \frac{\eta_2}{2\alpha + \eta_2}\left(\mathbf{X} + \frac{\Lambda_2}{\eta_2}\right)\right) - \mathbf{Z}\right\|_{\mathrm{F}}^2 \tag{36}$$

where $\phi(\cdot)$ can be any prior information; as an image in the feature space, the row of $\mathbf{Z}$ has the same characteristics as the original image, i.e., self-similarity. The above subproblem can be efficiently handled within a plug-and-play (PnP) framework by employing a suitable denoising operator as the corresponding proximal mapping. Therefore, based on the above formulation, the solution can be expressed as

$$\Psi_{\varepsilon\phi}(\mathbf{V}) = \arg\min_{\mathbf{Z}} \varepsilon\phi(\mathbf{Z}) + \frac{1}{2}\|\mathbf{V} - \mathbf{Z}\|_F^2 \tag{37}$$

where $\mathbf{V} = \mathbf{E}^T(2\alpha/(2\alpha + \eta_2)\mathbf{X}_{net} + \eta_2/(2\alpha + \eta_2)(\mathbf{X} + \Lambda_2/\eta_2))$ and $\varepsilon = \alpha\beta/(2\alpha + \eta_2)$. The above subproblem can then be solved by applying a denoising operator within the PnP framework, through which the variable $\mathbf{Z}$ is updated accordingly.

IV. Experiment

A. Experimental Setting

1) Compared Methods: To ensure the rigor of the experiment, this article fully considers each existing algorithm when selecting the comparison algorithms and finally selects the BDSD [38], BDSD-PC [38], and GS algorithms [8] in the category of CS methods; GLP-FS [39] and GLP-HPM-H algorithms [40] in the category of MRA methods; SR-D [41] and UCL [46] algorithms in the category of VO methods; and the PNN [42], PanNet [18], FusionNet [43], VONet [44],

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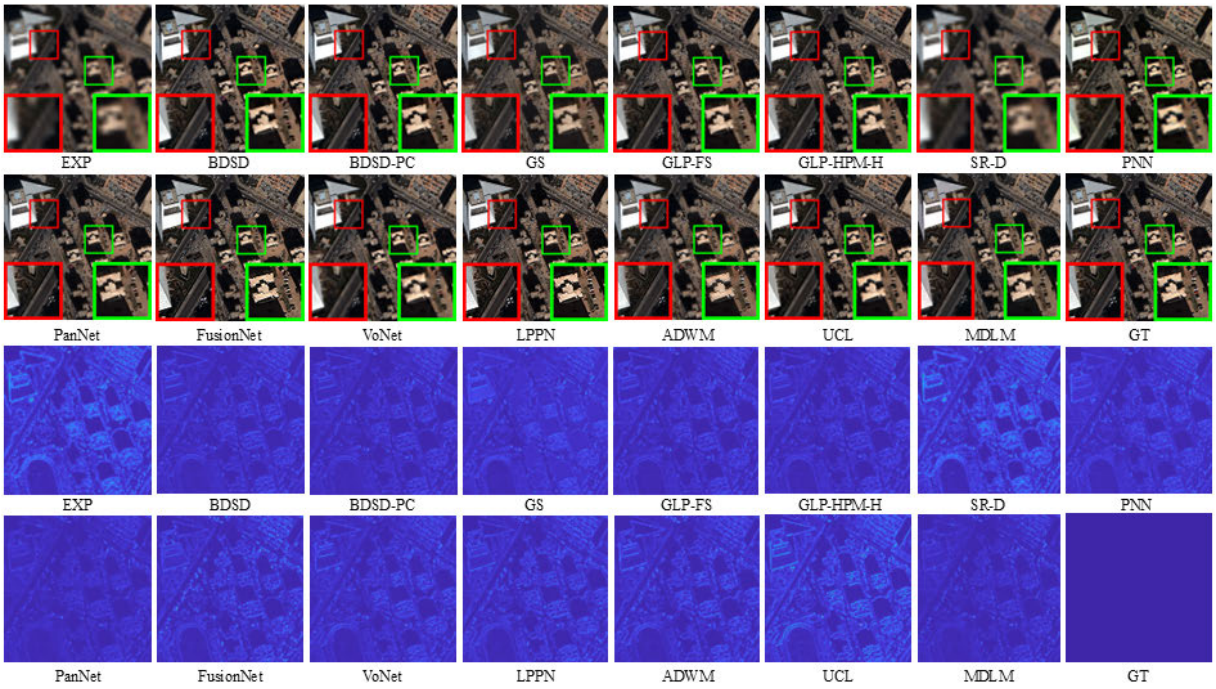


Fig. 3. Visual results on the reduced-resolution WV3 dataset. The scale of the PAN image is 64×64. (Top united row) Close-ups of the images fused by EXP [37], BDSD [38], BDSD-PC [38], GS [8], GLP-FS [39], GLP-HPM-H [40], SR-D [41], PNN [42], PanNet [18], FusionNet [43], VONet [44], VOGTNet [24], ADWM [45], UCL [46], and proposed algorithm and of the GT image. (Bottom united row) Corresponding residual maps using the GT image as a reference.

VOGTNet [24], and ADWM [45] algorithms in the category of DL methods. EXP [37] represents the resulting plot of direct linear interpolation upsampling MS into PAN images. For evaluation, we use the following metrics: spectral angle mapper (SAM) [47], Erreur Relative Globale Adimensionnelle de SynthÃse (ERGAS) [48], spatial correlation coefficient (SCC) [49], Qave (the average of the Q metric across spectral bands) and Q2n [50], as well as the quality metric without reference (QNR) [51].

2) Datasets: The dataset is divided into two categories based on the number of spectral bands: eight-band satellite imagery and four-band satellite imagery. The eight-band dataset includes WorldView-3 (WV3) and WorldView-2 (WV2) [52]. The WV3 sensor captures MS images in eight spectral bands and PAN images with a spatial resolution of 0.31 m, offering exceptional observational capability and fast data transmission. The WV2 satellite provides PAN images with a resolution of 0.5 m and MS images with a resolution of 1.8 m. The four-band dataset includes QuickBird (QB) [52] and IKONOS (IK). QB, which uses the Global Imaging System 2000, captures MS images with a resolution of 2.44 m and PAN images with a spatial resolution of 0.61 m. The IK satellite provides PAN images with a 1-m resolution and MS images with a 4-m resolution. The Gaofen-2 (GF-2) satellite is equipped with two PAN and MS sensors (PMSs). It provides high-resolution data comprising 0.81-m PAN images and 3.24-m MS images with four spectral bands (blue, green, red, and NIR), widely used for remote sensing applications. Specifically, we select three distinct scenes from the GF-2 data (denoted as GF2-1, GF2-2, and GF2-3) to comprehensively evaluate the performance in complex land cover scenarios. We test the datasets at both reduced resolution and full resolution, resulting in a total of 11 test datasets: WV3-RR, WV3-FR, WV2-RR, WV2-FR, QB-RR, QB-FR, IK-RR, IK-FR, and three GF2-RR datasets (GF2-1, GF2-2, and GF2-3).

3) Implementation Details: All the experiments are implemented in MATLAB on a computer equipped with 16 GB of RAM and an Intel[1] Core[2] i7-5960X CPU running at 3.00 GHz. The WV3 dataset is trained with PAN, MS, and GT image sizes of 64×64, $16 \times 16 \times 8$, and $64 \times 64 \times 8$, respectively. A model trained on the WV3 dataset is then used for both WV3 and WV2 (to test the generalization performance). The QB dataset is trained with PAN, MS, and GT image sizes of 64×64, $16 \times 16 \times 4$, and $64 \times 64 \times 4$, respectively. This process results in a QB model, which is used for QB and IK (to test generalization performance) evaluations.

B. Experimental Results and Analysis

1) Experimental Analysis of WV3: Fig. 3 shows the results of the reduced-resolution experiments. Since GT reference images are available, we subtract the GT image from the processed satellite images to generate greyscale images. The EXP image exhibits significant blurring, and all greyscale images contain numerous white spots. Among the various algorithms, BDSD, GS, and others preserve the original image details well but introduce some spectral distortion. The DL algorithms, on the other hand, produce slight blurring, particularly in terms of edge information. Among the MRA methods, the

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[2]Trademarked.

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TABLE I
QUANTITATIVE RESULTS ON THE REDUCED-RESOLUTION WV3 DATASET

Method	SAM	ERGAS	SCC	Q8	Qave
EXP [37]	6.101 ± 1.704	7.445 ± 1.852	0.735 ± 0.125	0.691 ± 0.130	0.695 ± 0.161
BDSD [38]	5.613 ± 1.868	4.137 ± 1.387	0.904 ± 0.059	0.799 ± 0.116	0.867 ± 0.156
BDSD-PC [39]	4.993 ± 0.659	3.989 ± 0.894	0.924 ± 0.062	0.889 ± 0.115	0.877 ± 0.146
GS [8]	6.135 ± 1.773	4.893 ± 0.543	0.816 ± 0.048	0.773 ± 0.119	0.776 ± 0.149
GLP-FS [40]	5.517 ± 1.301	4.147 ± 1.894	0.931 ± 0.071	0.898 ± 0.121	0.908 ± 0.135
GLP-HPM-H [41]	5.371 ± 1.834	4.148 ± 0.387	0.920 ± 0.075	0.867 ± 0.112	0.840 ± 0.143
SR-D [42]	5.500 ± 1.568	4.386 ± 1.923	0.939 ± 0.069	0.904 ± 0.129	0.904 ± 0.115
PNN [43]	5.383 ± 1.499	3.920 ± 0.857	0.922 ± 0.061	0.886 ± 0.111	0.889 ± 0.146
PanNet [18]	4.314 ± 0.506	2.997 ± 1.626	0.948 ± 0.064	0.896 ± 0.117	0.897 ± 0.116
FusionNet [44]	4.236 ± 0.557	3.343 ± 0.466	0.962 ± 0.054	0.894 ± 0.112	0.897 ± 0.124
VONet [45]	3.940 ± 0.540	2.833 ± 0.743	0.934 ± 0.053	0.901 ± 0.120	0.902 ± 0.123
VOGTNet [24]	3.79 ± 0.382	2.647 ± 0.245	0.967 ± 0.052	0.909 ± 0.118	0.910 ± 0.124
ADWM [46]	5.012 ± 0.673	5.459 ± 0.784	0.926 ± 0.053	0.723 ± 0.121	0.735 ± 0.105
UCL [47]	4.757 ± 0.596	3.937 ± 0.489	0.943 ± 0.042	0.824 ± 0.134	0.838 ± 0.126
MDLM	**3.791 ± 0.404**	**2.519 ± 0.157**	**0.977 ± 0.046**	**0.917 ± 0.107**	**0.916 ± 0.111**
Ideal Value	0	0	1	1	1

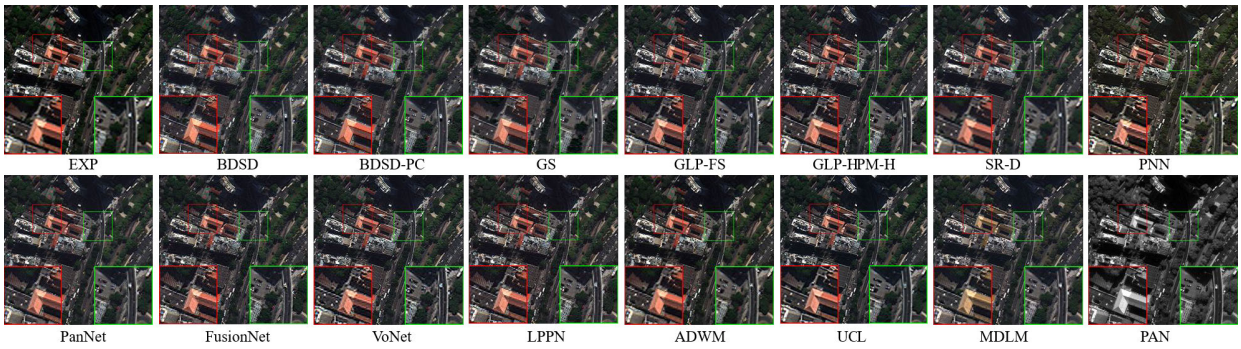


Fig. 4. Compositions for the full-resolution WV3 dataset. The scale of the PAN image is 64×64. Close-ups are shown in the lower left and lower right corners, zoomed-in view in to highlight the details.

greyscale image from GLP-HPM-H performs better than the other methods do. The method proposed in this article yields the best fusion quality, with fewer white spots in the greyscale image. From Table I, it is evident that in the reduced-resolution experiments, owing to the existence of reference image GT, we use reference-based evaluation metrics: SAM, ERGAS, SCC, Q8/4, and Qave. The results show that, after applying the CS algorithm, the spectral distortion is more pronounced. Among the CS-based methods, BDSD-PC performs better, yielding better performance in terms of band error, spatial correlation, and overall image quality than the other two algorithms do. Among the MRA methods, the fusion quality of both methods is not as good as that of BDSD-PC. For the optimization-based and DL methods, VOGTNet achieves indicators closer to the ideal values than most classical algorithms, while the newly added UCL method also shows relatively stable performance, especially in spatial correlation and overall quality; however, ADWM is less competitive in terms of ERGAS and overall image quality. The method proposed in this article better preserves spectral quality and achieves the best or comparable results in terms of spatial fidelity and overall image quality.

Fig. 4 shows the full-resolution results. Since there is no GT reference, the image detail information is compared with the PAN image only. From the figure, it is clear that the image obtained by BDSD-PC exhibits noticeable color differences compared with the others, indicating some spectral distortion.

TABLE II
QUANTITATIVE RESULTS ON THE FULL-RESOLUTION WV3 DATASET

Method	QNR	D_λ	D_s
EXP [37]	0.877 ± 0.055	0.006 ± 0.004	0.166 ± 0.033
BDSD [38]	0.838 ± 0.059	0.027 ± 0.011	0.115 ± 0.018
BDSD-PC [39]	0.817 ± 0.028	0.025 ± 0.010	0.174 ± 0.038
GS [8]	0.884 ± 0.037	0.036 ± 0.017	0.158 ± 0.017
GLP-FS [40]	0.886 ± 0.063	0.073 ± 0.021	0.157 ± 0.032
GLP-HPM-H [41]	0.824 ± 0.054	0.087 ± 0.048	0.194 ± 0.015
SR-D [42]	0.904 ± 0.031	0.029 ± 0.010	0.075 ± 0.028
PNN [43]	0.812 ± 0.032	0.059 ± 0.008	0.049 ± 0.029
PanNet [18]	0.911 ± 0.055	0.079 ± 0.023	0.118 ± 0.014
FusionNet [44]	0.924 ± 0.034	0.039 ± 0.014	0.064 ± 0.024
VONet [45]	0.949 ± 0.054	0.044 ± 0.007	0.048 ± 0.012
VOGTNet [24]	0.957 ± 0.031	0.031 ± 0.016	0.041 ± 0.017
ADWM [46]	0.874 ± 0.030	0.030 ± 0.003	0.099 ± 0.021
UCL [47]	0.932 ± 0.028	0.033 ± 0.017	0.037 ± 0.027
MDLM	**0.962 ± 0.023**	**0.023 ± 0.012**	**0.036 ± 0.011**
Ideal Value	1	0	0

However, its spatial detail is relatively intact. In contrast, the images from the SR-D, PNN, and other methods are blurred, and the spatial details are not fully preserved. Our proposed method maintains both the image details and edge information, preserves the color clarity, and achieves better fusion quality than traditional methods do. For the newly added ADWM and UCL methods, ADWM enhances some local spatial structures but tends to introduce spectral deviations and residual artifacts,

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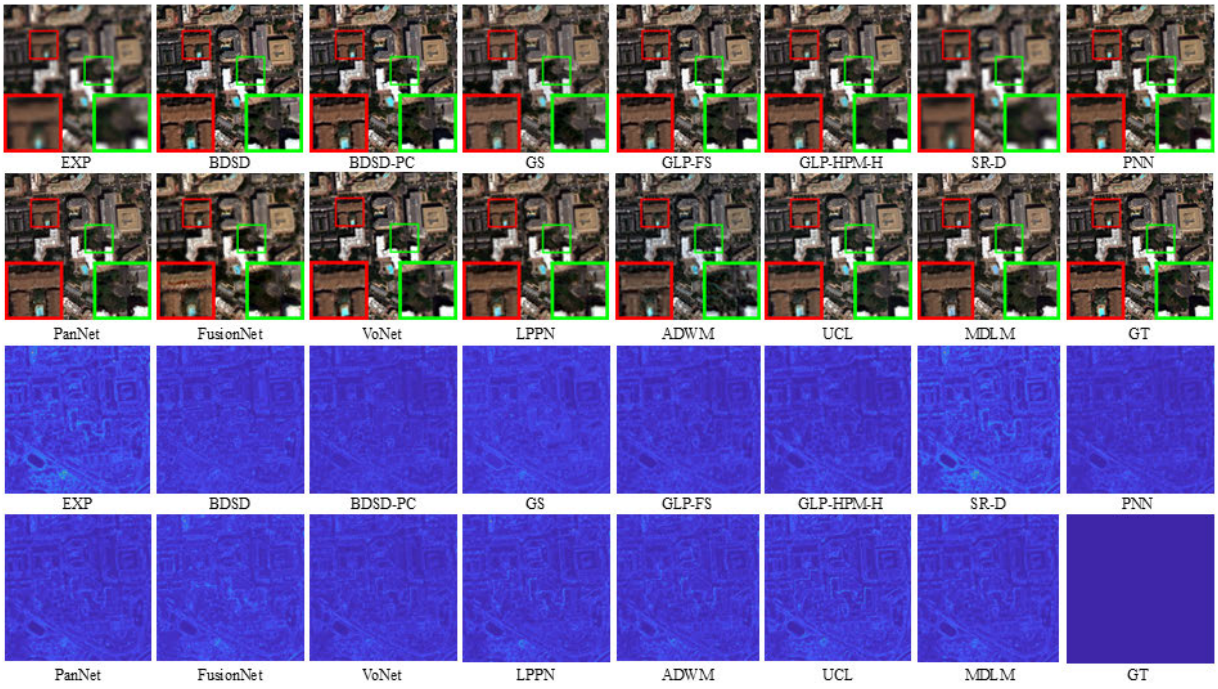


Fig. 5. Visual results on the reduced-resolution WV2 dataset. The scale of the PAN image is 64 × 64. (Top united row) Close-ups of images fused by EXP [37], BDSD [38], BDSD-PC [38], GS [8], GLP-FS [39], GLP-HPM-H [40], SR-D [41], PNN [42], PanNet [18], FusionNet [43], VONet [44], VOGTNet [24], ADWM [45], UCL [46], and proposed, and of the GT image. (Bottom united row) Corresponding residual maps using the GT image as a reference.

TABLE III
QUANTITATIVE RESULTS ON THE REDUCED-RESOLUTION WV2 DATASET

Method	SAM	ERGAS	SCC	Q8	Qave
EXP [37]	8.378 ± 1.902	7.355 ± 2.552	0.732 ± 0.128	0.675 ± 0.140	0.687 ± 0.167
BDSD [38]	8.805 ± 1.785	4.971 ± 2.402	0.888 ± 0.055	0.893 ± 0.096	0.881 ± 0.127
BDSD-PC [39]	7.789 ± 1.759	4.326 ± 1.611	0.935 ± 0.061	0.903 ± 0.092	0.909 ± 0.141
GS [8]	8.105 ± 1.622	5.455 ± 1.766	0.819 ± 0.038	0.837 ± 0.149	0.836 ± 0.138
GLP-FS [40]	7.994 ± 1.366	4.602 ± 3.117	0.880 ± 0.081	0.895 ± 0.101	0.889 ± 0.114
GLP-HPM-H [41]	7.443 ± 1.956	4.589 ± 1.543	0.839 ± 0.098	0.897 ± 0.083	0.896 ± 0.123
SR-D [42]	7.755 ± 1.566	4.715 ± 2.678	0.898 ± 0.061	0.888 ± 0.133	0.889 ± 0.110
PNN [43]	7.598 ± 1.489	5.076 ± 1.969	0.906 ± 0.058	0.873 ± 0.134	0.877 ± 0.121
PanNet [18]	7.721 ± 1.509	4.959 ± 2.639	0.918 ± 0.059	0.877 ± 0.127	0.872 ± 0.122
FusionNet [44]	7.749 ± 1.312	4.803 ± 1.798	0.883 ± 0.054	0.854 ± 0.093	0.852 ± 0.122
VONet [45]	6.240 ± 1.720	3.637 ± 1.847	0.945 ± 0.050	0.902 ± 0.112	0.902 ± 0.135
VOGTNet [24]	6.517 ± 1.636	4.103 ± 1.236	0.932 ± 0.067	0.875 ± 0.113	0.879 ± 0.147
ADWM [46]	6.421 ± 0.691	5.532 ± 1.226	0.860 ± 0.006	0.768 ± 0.037	0.770 ± 0.086
UCL [47]	5.626 ± 1.235	3.980 ± 1.617	0.929 ± 0.045	0.858 ± 0.035	0.854 ± 0.041
MDLM	6.040 ± 1.442	$\mathbf{3.100 \pm 0.925}$	$\mathbf{0.961 \pm 0.055}$	$\mathbf{0.913 \pm 0.136}$	$\mathbf{0.916 \pm 0.119}$
Ideal Value	0	0	1	1	1

whereas UCL shows relatively stable spectral preservation but still loses some fine edge and texture details. In comparison, the proposed method provides a better balance between spatial detail recovery and spectral fidelity. From Table II, it is evident that in the full-resolution experiments, where no GT reference image is available, we use the no-reference quality metric QNR. Unlike in the reduced-resolution experiments, the QNR metric for BDSD-PC in the CS algorithm is the least ideal, showing significant spatial and spectral distortion. DL and optimization-based methods generally outperform MRA methods, with QNR values closer to 1, indicating smaller spatial and spectral distortion. Among the newly added comparison methods, UCL obtains a relatively competitive QNR value, whereas ADWM shows less stable performance in the no-reference quality assessment. Compared with the other algorithms, the proposed method achieves the highest QNR value and maintains a better balance between spectral and spatial distortions, as reflected by its competitive D_λ and D_s values.

2) Experimental Analysis of WV2: as detailed as BDSD-PC. SR-D, which is based on the VONet algorithm, shows some improvement in both spectral and spatial resolution, but it falls short of the performance of several DL methods. Among the DL approaches, PanNet produces the smallest errors. However, color distortion is visible in the FusionNet image, particularly in the lower right area, where details are blurred. For the newly added ADWM and UCL methods, ADWM preserves some spatial structures in the fused

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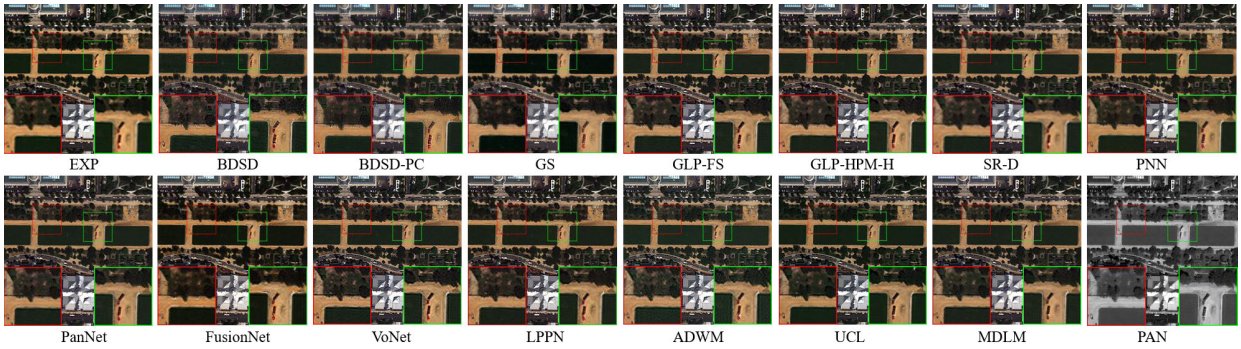


Fig. 6. Compositions for the full-resolution WV2 dataset. The scale of the PAN image is 64 × 64. Close-ups are shown in the lower left and lower right corners, zoomed-in view in to highlight the details.

image, but its residual map still contains noticeable structural traces, indicating that part of the spatial and spectral errors is not fully suppressed. UCL presents relatively stable visual quality, but some residual edges and texture differences remain in the enlarged regions. In comparison, the algorithm proposed in this article produces a more uniform residual map and preserves image edges more effectively, demonstrating the superior generalization performance of our trained network. From Table III, it is evident that BDSD-PC outperforms the other methods across the evaluation metrics for reduced-resolution cases. Among the MRA algorithms, the GLP-HPM-H method shows better performance, while the VOGTNet method yields promising numerical results compared with other DL approaches. UCL obtains a competitive SAM value in the reduced-resolution experiment, indicating favorable spectral angle preservation, but it is still inferior to the proposed method in terms of ERGAS, SCC, Q8, and Qave. In contrast, ADWM shows less competitive performance, especially in terms of ERGAS and overall image quality. Furthermore, our proposed algorithm achieves the best or highly competitive results across various indicators, demonstrating its ability to effectively enhance the spatial detail quality of the fused image while minimizing spectral distortion. In the **reduced-resolution** experiments (Fig. 5), from the greyscale display, it is observed that BDSD improves spatial quality compared with GS, although it is not

In the **full-resolution** experiments (Fig. 6), the details in the BDSD and GS images are not sufficiently clear. The improvements in the GLP-HPM-H structure are clearly visible. On the other hand, FusionNet produces significant color distortion in the lower left area, with spectral values showing poor correspondence with the original image. Compared with ADWM and UCL, our method shows clearer spatial improvements and better detail recovery while maintaining more consistent color information, providing the best results overall. From Table IV, it is evident that the no-reference quantitative results further confirm the effectiveness of the proposed method. Compared with traditional CS and MRA methods, DL and optimization-based methods generally obtain better QNR values, indicating improved spatial and spectral consistency. Among the newly added comparison methods, ADWM and UCL show less stable performance in the full-resolution quality assessment. In contrast, the proposed algorithm achieves the best or highly competitive results, demonstrating that our framework yields competitive performance and effectively improves both the spatial and spectral aspects of image fusion.

TABLE IV

QUANTITATIVE RESULTS ON THE FULL-RESOLUTION WV2 DATASET

Method	QNR	D_λ	D_s
EXP [37]	0.869 ± 0.065	0.007 ± 0.002	0.180 ± 0.036
BDSD [38]	0.825 ± 0.052	0.031 ± 0.014	0.125 ± 0.024
BDSD-PC [39]	0.792 ± 0.017	0.028 ± 0.010	0.192 ± 0.047
GS [8]	0.889 ± 0.043	0.037 ± 0.018	0.167 ± 0.027
GLP-FS [40]	0.897 ± 0.064	0.084 ± 0.024	0.187 ± 0.026
GLP-HPM-H [41]	0.827 ± 0.055	0.081 ± 0.038	0.189 ± 0.007
SR-D [42]	0.899 ± 0.039	0.033 ± 0.008	0.064 ± 0.032
PNN [43]	0.821 ± 0.044	0.047 ± 0.016	0.039 ± 0.020
PanNet [18]	0.920 ± 0.052	0.093 ± 0.024	0.105 ± 0.015
FusionNet [44]	0.911 ± 0.042	0.044 ± 0.014	0.067 ± 0.025
VONet [45]	0.957 ± 0.053	0.043 ± 0.006	0.053 ± 0.040
VOGTNet [24]	0.952 ± 0.043	0.040 ± 0.014	0.058 ± 0.021
ADWM [46]	0.814 ± 0.011	0.089 ± 0.055	0.106 ± 0.041
UCL [47]	0.776 ± 0.024	0.126 ± 0.153	0.115 ± 0.096
MDLM	**0.965 ± 0.037**	**0.034 ± 0.009**	**0.045 ± 0.009**
Ideal Value	1	0	0

3) Experimental Analysis of QB: In the **reduced-resolution** experiments (Fig. 7), we observe that the vegetation areas in the GS, SR-D, PanNet, and VOGTNet methods exhibit noticeable spatial distortion and a loss of fine details. The PNN method shows some spatial improvement, but with significant color distortions. ADWM and UCL methods show relatively stable visual results in the reduced-resolution experiments, but local blurring and residual detail differences can still be observed. In contrast, the experimental results of the proposed method maintain good color retention and demonstrate superior spatial enhancement.

From Table V, it is evident that the proposed method achieves the best or highly competitive results across various reference-based evaluation metrics. Specifically, MDLM obtains the best SAM and SCC values, indicating that it can effectively reduce spectral distortion while improving spatial correlation. Although UCL achieves a slightly better ERGAS value, the proposed method shows better overall performance in terms of spectral preservation, spatial fidelity, and image quality. These results demonstrate that the proposed method can effectively enhance spatial details while maintaining good spectral consistency on the QB dataset. In the **full-resolution**

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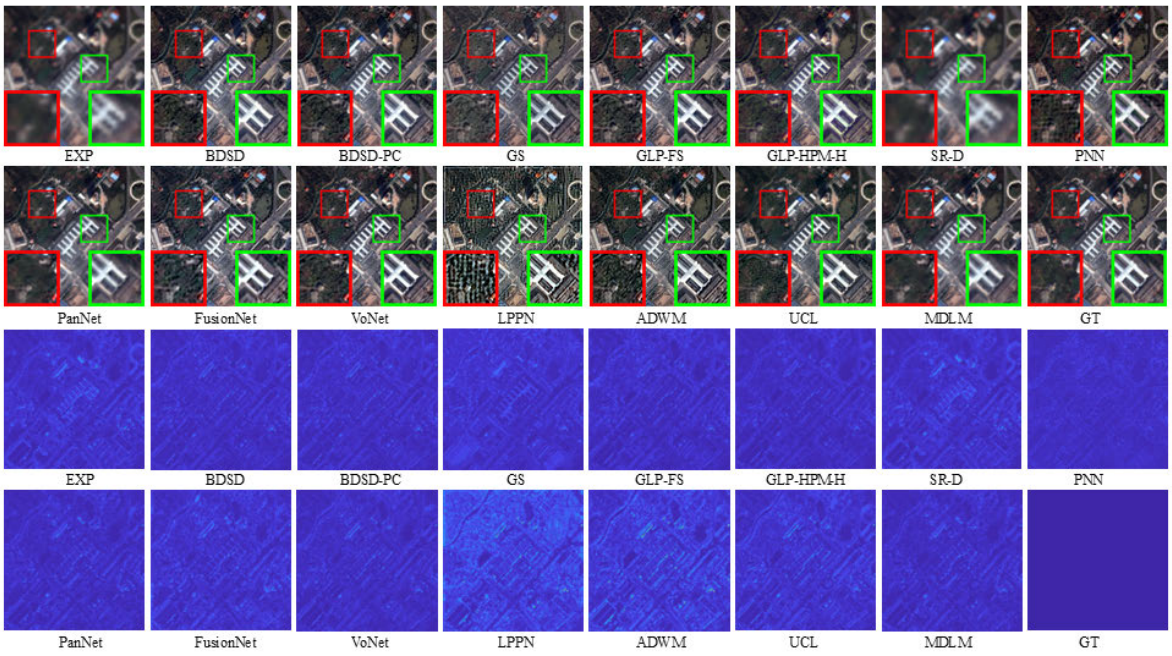


Fig. 7. Visual results on the reduced-resolution QB dataset. The scale of the PAN image is 64 × 64. (Top united row) Close-ups of images fused by EXP [37], BDSD [38], BDSD-PC [38], GS [8], GLP-FS [39], GLP-HPM-H [40], SR-D [41], PNN [42], PanNet [18], FusionNet [43], VONet [44], VOGTNet [24], ADWM [45], UCL [46], and proposed, and of the GT image. (Bottom united row) Corresponding residual maps using the GT image as a reference.

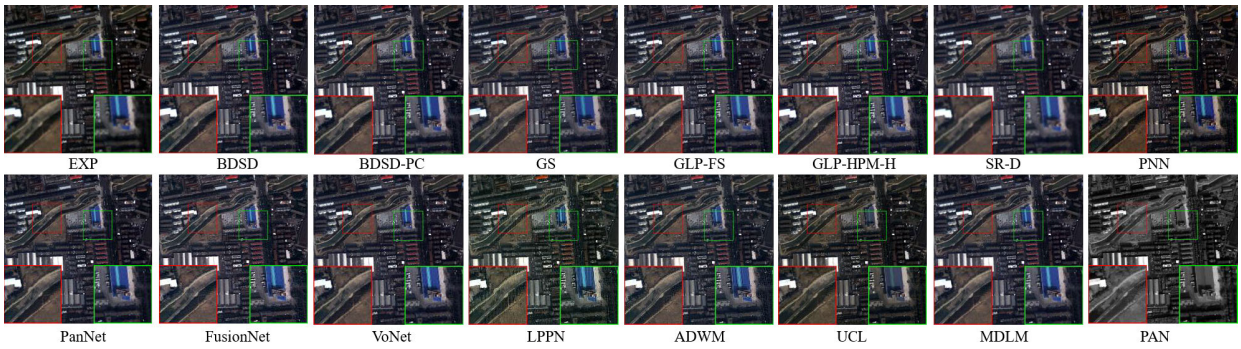


Fig. 8. Compositions for the full-resolution QB dataset. The scale of the PAN image is 64 × 64. Close-ups are shown in the lower left and lower right corners, zoomed-in view in to highlight the details.

TABLE V

QUANTITATIVE RESULTS ON THE REDUCED-RESOLUTION QB DATASET

Method	SAM	ERGAS	SCC	Q4	Qave
EXP [37]	4.769 ± 0.144	9.082 ± 0.188	0.724 ± 0.141	0.717 ± 0.013	0.656 ± 0.141
BDSD [38]	5.213 ± 0.126	6.205 ± 0.149	0.854 ± 0.079	0.824 ± 0.008	0.777 ± 0.135
BDSD-PC [39]	4.232 ± 0.134	5.168 ± 0.129	0.870 ± 0.073	0.945 ± 0.004	0.701 ± 0.129
GS [8]	4.735 ± 0.147	6.116 ± 0.169	0.849 ± 0.051	0.845 ± 0.019	0.767 ± 0.146
GLP-FS [40]	4.237 ± 0.174	5.419 ± 0.180	0.865 ± 0.089	0.822 ± 0.001	0.713 ± 0.150
GLP-HPM-H [41]	4.222 ± 0.074	6.505 ± 0.199	0.860 ± 0.078	0.909 ± 0.002	0.795 ± 0.095
SR-D [42]	4.216 ± 0.085	6.376 ± 0.030	0.843 ± 0.049	0.842 ± 0.002	0.772 ± 0.150
PNN [43]	4.083 ± 0.089	5.299 ± 0.177	0.812 ± 0.045	0.810 ± 0.003	0.816 ± 0.095
PanNet [18]	3.314 ± 0.092	4.987 ± 0.134	0.855 ± 0.079	0.837 ± 0.013	0.773 ± 0.150
FusionNet [44]	3.536 ± 0.157	4.638 ± 0.199	0.845 ± 0.053	0.823 ± 0.012	0.767 ± 0.152
VONet [45]	3.340 ± 0.045	4.554 ± 0.120	0.909 ± 0.067	0.939 ± 0.017	0.802 ± 0.152
VOGTNet [24]	3.579 ± 0.068	4.347 ± 0.116	0.903 ± 0.041	0.938 ± 0.019	0.834 ± 0.089
ADWM [46]	4.640 ± 0.067	4.945 ± 0.235	0.900 ± 0.104	0.889 ± 0.009	0.899 ± 0.119
UCL [47]	4.422 ± 0.458	4.161 ± 0.016	0.910 ± 0.041	0.821 ± 0.019	0.834 ± 0.097
MDLM	$\mathbf{3.271 \pm 0.059}$	4.219 ± 0.089	$\mathbf{0.915 \pm 0.033}$	$\mathbf{0.964 \pm 0.018}$	$\mathbf{0.855 \pm 0.078}$
Ideal Value	0	0	1	1	1

experiments (Fig. 8), the SR-D and PNN methods show notable spatial improvements over BDSD and GS, but they suffer from significant color shifts. From the four close-ups, it is evident that while buildings, roads, and vegetation are

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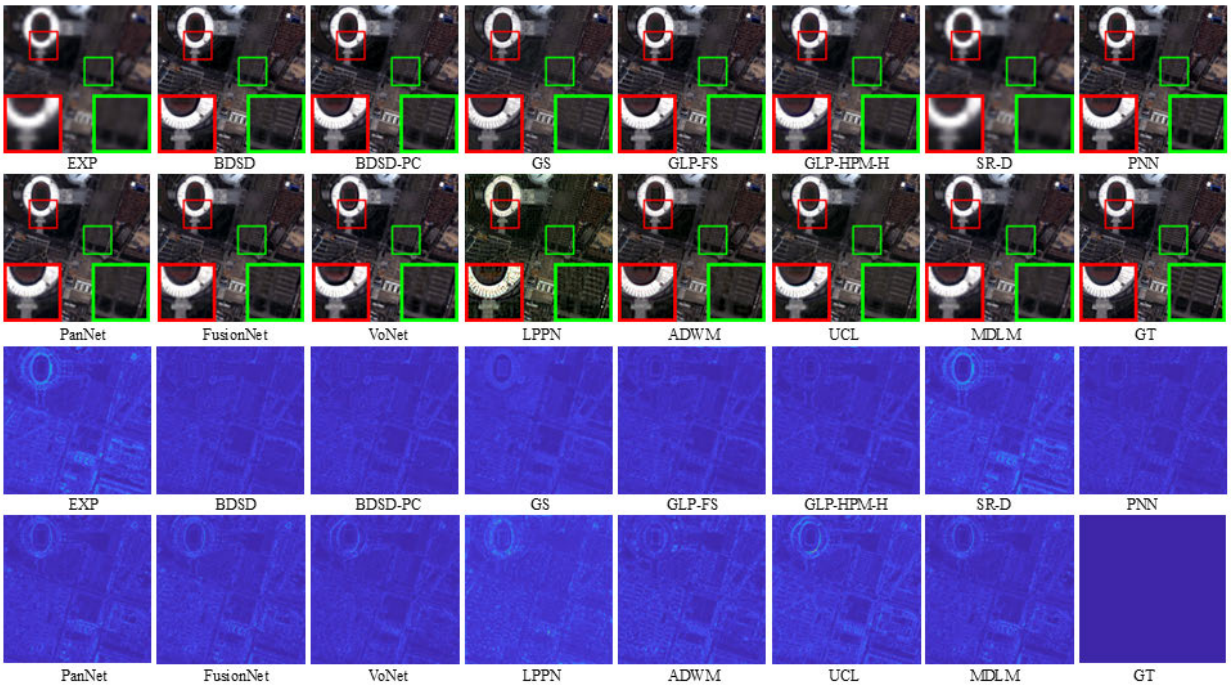


Fig. 9. Visual results on the reduced-resolution IK dataset. The scale of the PAN image is 64×64. (Top united row) Close-ups of images fused by EXP [37], BDSD [38], BDSD-PC [38], GS [8], GLP-FS [39], GLP-HPM-H [40], SR-D [41], PNN [42], PanNet [18], FusionNet [43], VONet [44], VOGTNet [24], ADWM [45], UCL [46], and proposed, and of the GT image. (Bottom united row) corresponding residual maps using the GT image as a reference.

TABLE VI
QUANTITATIVE RESULTS ON THE FULL-RESOLUTION QB DATASET

Method	QNR	D_λ	D_s
EXP [37]	0.853 ± 0.045	0.004 ± 0.002	0.174 ± 0.026
BDSD [38]	0.823 ± 0.053	0.031 ± 0.012	0.124 ± 0.022
BDSD-PC [39]	0.802 ± 0.011	0.026 ± 0.007	0.179 ± 0.030
GS [8]	0.858 ± 0.037	0.038 ± 0.012	0.164 ± 0.021
GLP-FS [40]	0.856 ± 0.054	0.069 ± 0.024	0.167 ± 0.032
GLP-HPM-H [41]	0.804 ± 0.048	0.097 ± 0.050	0.198 ± 0.006
SR-D [42]	0.885 ± 0.011	0.036 ± 0.012	0.077 ± 0.022
PNN [43]	0.805 ± 0.028	0.069 ± 0.010	0.058 ± 0.027
PanNet [18]	0.915 ± 0.050	0.050 ± 0.022	0.124 ± 0.019
FusionNet [44]	0.927 ± 0.016	0.049 ± 0.013	0.067 ± 0.029
VONet [45]	0.935 ± 0.018	0.037 ± 0.010	0.058 ± 0.014
VOGTNet [24]	0.932 ± 0.021	0.033 ± 0.018	0.054 ± 0.013
ADWM [46]	0.813 ± 0.064	0.113 ± 0.007	0.196 ± 0.065
UCL [47]	0.876 ± 0.029	0.095 ± 0.004	0.094 ± 0.030
MDLM	**0.943 ± 0.015**	**0.027 ± 0.015**	**0.049 ± 0.014**
Ideal Value	1	0	0

clearly visible in the PAN image, they appear very blurry in the EXP image. The experimental results of our proposed method show high visual quality without visible artifacts or spectral distortions. From Table VI, it can be observed that the proposed method achieves the best QNR value and maintains competitive D_λ and D_s values, indicating a better balance between spectral and spatial distortions. Compared with the other algorithms, MDLM better preserves spatial and spectral details in the full-resolution experiments, further confirming the effectiveness of the proposed framework for QB image fusion.

4) Experimental Analysis of IK: For the **reduced-resolution** IK dataset, we choose to include sample images of buildings and roads, as shown in Fig. 9, to allow meaningful comparisons of pansharpening quality between competing techniques. In the low-resolution experiments, the EXP image is blurred overall compared with the reference image, and although the GS method improves spatial resolution, it introduces significant spectral distortion. The fusion results from PanNet, VOGTNet, UCL, ADWM, and VONet also show vague building regions. In contrast, the method proposed in this article has a good color retention effect and significantly improves spatial resolution.

As shown in Table VII, among the CS algorithms, both BDSD-PC and GS produce better results. Among the DL-based methods, VONet and VOGTNet outperform SR-D and PanNet in terms of numerical performance. UCL shows relatively stable quantitative performance, whereas ADWM is less competitive in terms of overall image quality. Our proposed method achieves the closest values to the ideal metrics, demonstrating the strong generalization ability of our trained network.

In the **full-resolution** experiments (Fig. 10), the samples include buildings, roads, and trees. The vegetation of the EXP and GS images has a large color difference, and the color retention effect of the other traditional methods is not as good. Although all methods yield spatial improvements, the structure of the image is less clear than that of the image produced by our method. In Table VIII, although VONet and FusionNet achieve relatively better results, UCL and ADWM still obtain lower no-reference quality than the proposed method. Our method consistently maintains superiority across all nonreference metrics, achieving the best performance overall, thus highlighting the effectiveness and practical utility of our approach.

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TABLE VII
QUANTITATIVE RESULTS ON THE REDUCED-RESOLUTION IK DATASET

Method	SAM	ERGAS	SCC	Q4	Qave
EXP [37]	8.260 ± 0.240	9.453 ± 0.349	0.718 ± 0.142	0.676 ± 0.021	0.597 ± 0.081
BDSD [38]	8.686 ± 0.276	7.042 ± 0.234	0.883 ± 0.070	0.733 ± 0.011	0.732 ± 0.120
BDSD-PC [39]	7.356 ± 0.257	6.276 ± 0.289	0.870 ± 0.071	0.903 ± 0.003	0.606 ± 0.123
GS [8]	7.103 ± 0.155	7.357 ± 0.365	0.858 ± 0.043	0.762 ± 0.022	0.695 ± 0.120
GLP-FS [40]	7.432 ± 0.181	6.023 ± 0.360	0.883 ± 0.081	0.746 ± 0.007	0.683 ± 0.161
GLP-HPM-H [41]	6.822 ± 0.227	7.133 ± 0.262	0.861 ± 0.084	0.881 ± 0.011	0.758 ± 0.089
SR-D [42]	7.484 ± 0.265	7.426 ± 0.135	0.865 ± 0.045	0.786 ± 0.009	0.683 ± 0.139
PNN [43]	6.508 ± 0.191	6.284 ± 0.282	0.826 ± 0.047	0.779 ± 0.004	0.727 ± 0.089
PanNet [18]	7.104 ± 0.105	6.378 ± 0.287	0.852 ± 0.089	0.773 ± 0.010	0.681 ± 0.157
FusionNet [44]	5.678 ± 0.321	5.437 ± 0.339	0.874 ± 0.052	0.847 ± 0.009	0.667 ± 0.131
VONet [45]	5.325 ± 0.106	4.989 ± 0.289	0.904 ± 0.031	0.912 ± 0.015	0.740 ± 0.077
VOGTNet [24]	5.483 ± 0.124	4.898 ± 0.225	0.910 ± 0.043	0.907 ± 0.011	0.798 ± 0.065
ADWM [46]	5.259 ± 0.036	4.933 ± 0.131	0.893 ± 0.006	0.868 ± 0.019	0.868 ± 0.088
UCL [47]	5.254 ± 0.458	4.917 ± 0.016	0.903 ± 0.041	0.848 ± 0.142	0.852 ± 0.103
MDLM	**5.233 ± 0.124**	**4.791 ± 0.170**	**0.911 ± 0.032**	**0.853 ± 0.016**	**0.868 ± 0.054**
Ideal Value	0	0	1	1	1

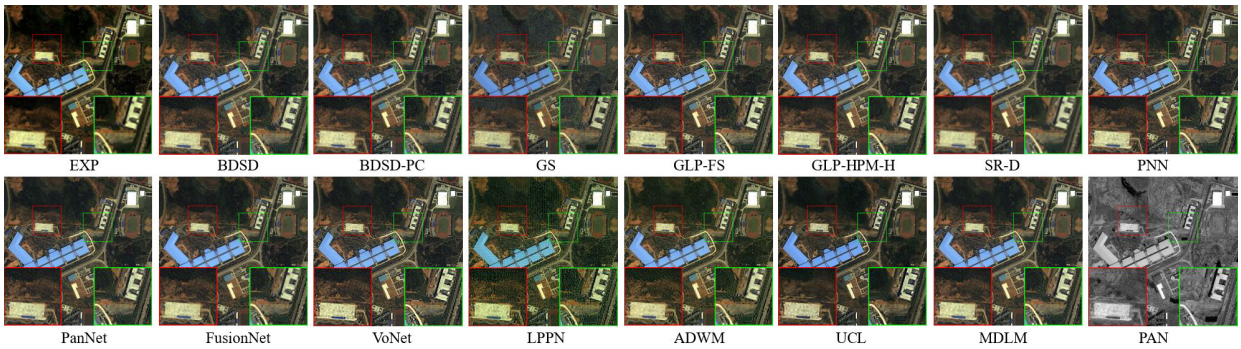


Fig. 10. Compositions for the full-resolution IK dataset. The scale of the PAN image is 64×64. Close-ups are shown in the lower left and lower right corners, zoomed-in view in to highlight the details.

TABLE VIII
QUANTITATIVE RESULTS ON THE FULL-RESOLUTION IK DATASET

Method	QNR	D_λ	D_s
EXP [37]	0.842 ± 0.058	0.005 ± 0.003	0.179 ± 0.023
BDSD [38]	0.825 ± 0.055	0.034 ± 0.008	0.118 ± 0.027
BDSD-PC [39]	0.777 ± 0.021	0.028 ± 0.003	0.161 ± 0.027
GS [8]	0.845 ± 0.012	0.035 ± 0.012	0.140 ± 0.020
GLP-FS [40]	0.863 ± 0.065	0.077 ± 0.029	0.163 ± 0.027
GLP-HPM-H [41]	0.806 ± 0.038	0.096 ± 0.048	0.182 ± 0.080
SR-D [42]	0.894 ± 0.009	0.039 ± 0.013	0.089 ± 0.018
PNN [43]	0.801 ± 0.019	0.072 ± 0.007	0.051 ± 0.022
PanNet [18]	0.886 ± 0.043	0.056 ± 0.025	0.141 ± 0.021
FusionNet [44]	0.931 ± 0.031	0.055 ± 0.010	0.079 ± 0.027
VONet [45]	0.943 ± 0.029	0.041 ± 0.010	0.062 ± 0.016
VOGTNet [24]	0.933 ± 0.021	0.034 ± 0.016	0.049 ± 0.014
ADWM [46]	0.770 ± 0.038	0.129 ± 0.005	0.114 ± 0.053
UCL [47]	0.785 ± 0.006	0.247 ± 0.041	0.232 ± 0.035
MDLM	**0.945 ± 0.015**	**0.030 ± 0.019**	**0.046 ± 0.013**
Ideal Value	1	0	0

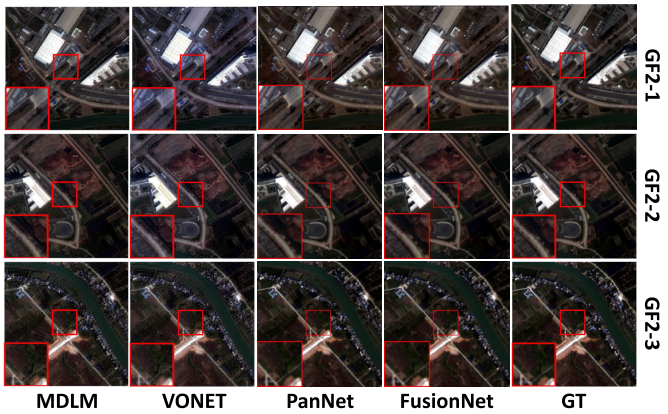


Fig. 11. Visual comparison of different fusion methods on three GF-2 test datasets.

5) Experimental Analysis of GF-2 Datasets: To comprehensively evaluate the generalization ability and robustness of the proposed MDLM method across different scenes, experiments were further conducted on three GF-2 datasets featuring urban structures and complex vegetation. Fig. 11 shows the visual comparison of different fusion methods on the three GF-2 test datasets. It can be observed that VONet, PanNet, and FusionNet can recover certain spatial structures, but some local blurring, edge degradation, and color deviations still exist in complex regions. In particular, the building contours, road boundaries, and vegetation textures are not sufficiently clear in some comparison results. In contrast, the proposed MDLM method produces clearer spatial structures and more natural color information, and its visual results are more consistent with the GT images. These visual comparisons indicate that the proposed method has better robustness in different GF-2 scenes and can effectively preserve both spatial details and

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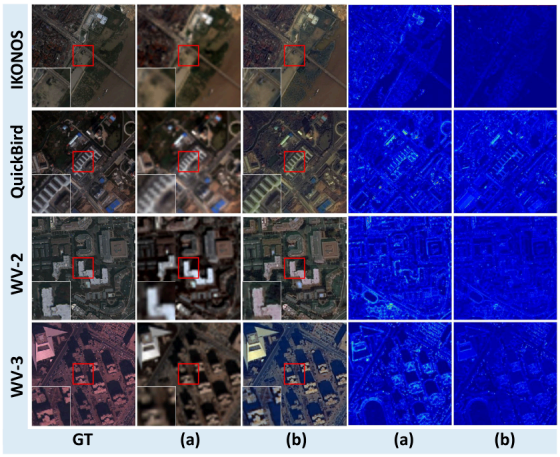


Fig. 12. Visual and quantitative comparison of estimation errors. (a) Direct pixel estimation: shows widespread pixel-wise errors due to the lack of structural constraints. (b) Proposed subspace optimization: the proposed method effectively suppresses the estimation errors, validating the correction capability of the subspace constraint.

TABLE IX
QUANTITATIVE COMPARISON RESULTS ON THREE GF-2 DATASETS

Dataset	Method	SAM ↓	ERGAS ↓	SCC ↑
GF2-1	MDLM	**1.1126**	**1.0664**	0.8399
	VONet	1.3488	3.5536	**0.9211**
	PanNet	1.8965	1.6770	0.8135
	FusionNet	1.8909	1.6681	0.8350
GF2-2	MDLM	**0.9982**	**1.0187**	0.8182
	VONet	1.1706	3.5452	**0.9009**
	PanNet	1.6092	1.5216	0.7838
	FusionNet	1.6092	1.7823	0.7220
GF2-3	MDLM	**1.1535**	**1.4625**	0.7672
	VONet	1.2229	3.6602	**0.8769**
	PanNet	1.7305	1.5816	0.7513
	FusionNet	1.8424	1.7551	0.7050

spectral information. Consistent with the previous settings, the evaluation was performed under the reduced-resolution protocol using SAM, ERGAS, and SCC metrics. As shown in Table IX, the proposed method consistently achieves better performance in terms of spectral fidelity, reflected by lower SAM and ERGAS values across all datasets. Meanwhile, although certain DL methods obtain slightly higher SCC values, MDLM maintains competitive spatial performance while significantly reducing spectral distortion. Overall, these results demonstrate that the proposed method effectively balances spatial detail preservation and spectral consistency, and exhibits strong robustness across different GF-2 scenes.

C. Ablation Analysis and Discussion

1) Pixel-Wise Error Analysis: To explicitly investigate the limitations of direct pixel-level estimation, we compare the *Direct Pixel Estimation* baseline with the proposed subspace optimization strategy presented in Fig. 12. As shown in Table X, the proposed method consistently achieves higher PSNR and lower ERGAS values across all four datasets, including IK, QB, WV2, and WV3. In contrast, the direct pixel estimation baseline produces relatively lower PSNR and higher ERGAS, indicating that directly predicting the fused image at the pixel level may lead to insufficient reconstruction accuracy and increased spectral distortion. By introducing the DS representation and the subsequent optimization process, the proposed method effectively improves the reconstruction quality while reducing global spectral errors. These results confirm that the subspace constraint is beneficial for alleviating the limitations of direct pixel-level estimation.

TABLE X
QUANTITATIVE COMPARISON OF DIRECT PIXEL EST. AND PROPOSED METHODS

Dataset	Direct Pixel Est. (Base)		Proposed (Subspace)	
	PSNR ↑	ERGAS ↓	PSNR ↑	ERGAS ↓
IKONOS	35.12	1.70	36.78	1.40
QB	36.51	2.28	37.34	2.12
WV2	23.89	7.36	25.91	5.64
WV3	22.86	7.45	33.14	5.35

TABLE XI
QUANTITATIVE COMPARISON RESULTS (MAE AND SAM) ON SEVEN DATASETS

Metric	Method	IKONOS	QB	WV2	WV3	GF2-1	GF2-2	GF2-3
MAE ↓	MDLM	6.313	8.192	29.557	20.617	19.211	15.980	6.975
	PanNet	7.324	8.065	29.180	19.071	19.247	16.008	6.803
SAM ↓	MDLM	1.3546	2.0898	7.7216	5.1654	1.7035	1.6222	1.6467
	PanNet	1.6953	2.0907	8.3912	4.9916	1.7082	1.6263	1.6479

2) Cross-Dataset Comparison With PanNet: To further evaluate the cross-dataset generalization ability of the proposed MDLM method, Table XI reports the quantitative comparison between MDLM and PanNet on seven datasets, including IK, QB, WV2, WV3, and three GF-2 scenes. MAE is used to measure pixel-level reconstruction error, while SAM is adopted to evaluate spectral distortion. As shown in Fig. 13, MDLM achieves lower SAM values than PanNet on most datasets, indicating better spectral preservation and more stable spectral consistency. In terms of MAE, MDLM obtains comparable results to PanNet, with better performance on IK and GF-2, while PanNet achieves slightly lower MAE on several other datasets. These results suggest that the proposed method is particularly effective in reducing spectral distortion while maintaining competitive pixel-level reconstruction accuracy across different datasets.

3) Parametric Analysis: Our model includes four parameters. Fig. 14 shows the variations in the SAM and the general image quality metric Q8, as these parameters are adjusted in the reduced-resolution experiment. To simplify the parameter selection process, we modify all the parameters except those under analysis. The results clearly indicate that the optimal combination is $\lambda = 10^{-4}$, $\alpha = 2.75 \times 10^{-3}$, $\eta_1 = 5 \times 10^{-2}$, and $\eta_2 = 10^{-2}$, based on the adopted quality metrics. This approach can be similarly applied to other remote sensing datasets to determine the corresponding optimal parameter configuration.

4) Convergence Analysis: As shown in the convergence curve, the objective value decreases rapidly during the early iterations and gradually becomes stable after approximately 60 iterations. This observation indicates that the proposed MDLM algorithm exhibits stable empirical convergence behavior.

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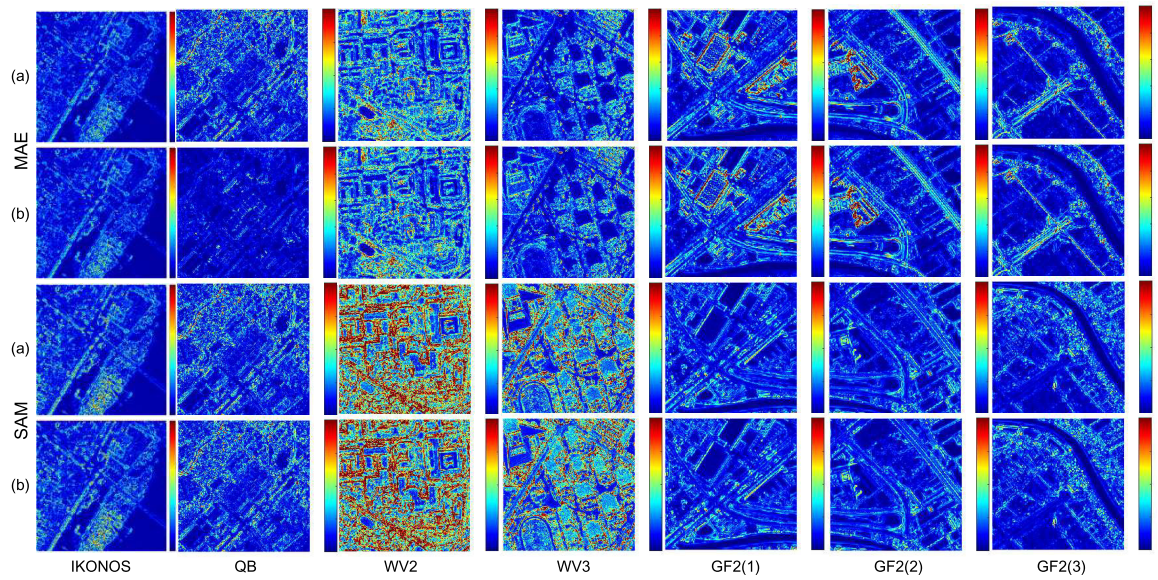


Fig. 13. Quantitative performance comparison between the proposed MDLM method and the baseline PanNet method across seven different satellite datasets. (a) Comparison of MAE values. (b) Comparison of SAM values. The datasets are arranged from left to right as: IK, QB, WV2, WV3, and three GF-2 datasets.

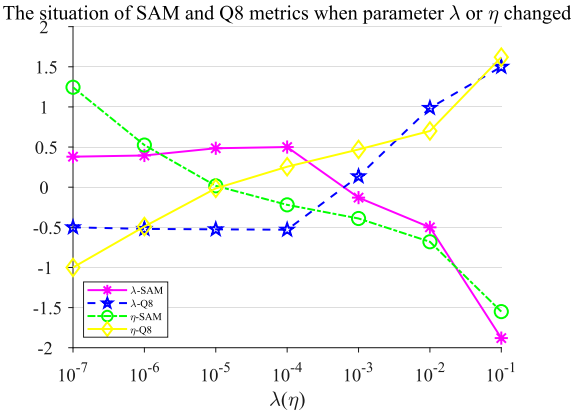


Fig. 14. Changes in metrics after fine-tuning different experimental parameters on the same dataset.

Therefore, setting the maximum number of iterations to 100 was sufficient to obtain stable results in our experiments.

V. Conclusion

In this article, we propose a novel fusion scheme for pansharpening from a DS perspective. We introduce the DS prior to achieve knowledge transfer via a deep neural network in an optimization model. Furthermore, the RDM model is exploited with a nonlocal similarity prior in the spatial fidelity term to reduce interference. Qualitative and quantitative results from both reduced-resolution and full-resolution experiments, which were conducted on several widely used datasets, demonstrate the effectiveness and competitiveness of the proposed framework. In future work, we will further investigate joint optimization strategies that bridge traditional optimization models and deep neural networks, so as to better exploit the strengths of both paradigms.

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